



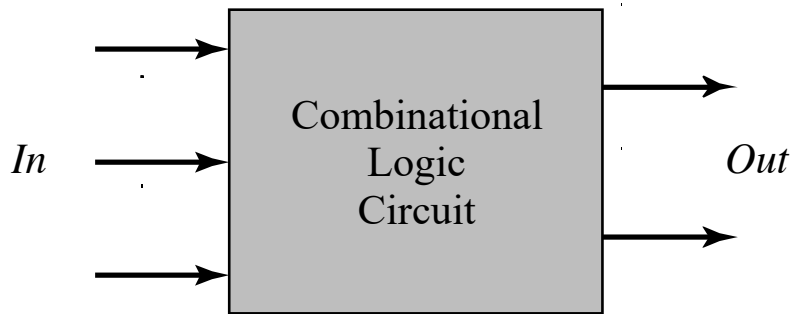
Digital Integrated Circuits

A Design Perspective

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Anantha Chandrakasan
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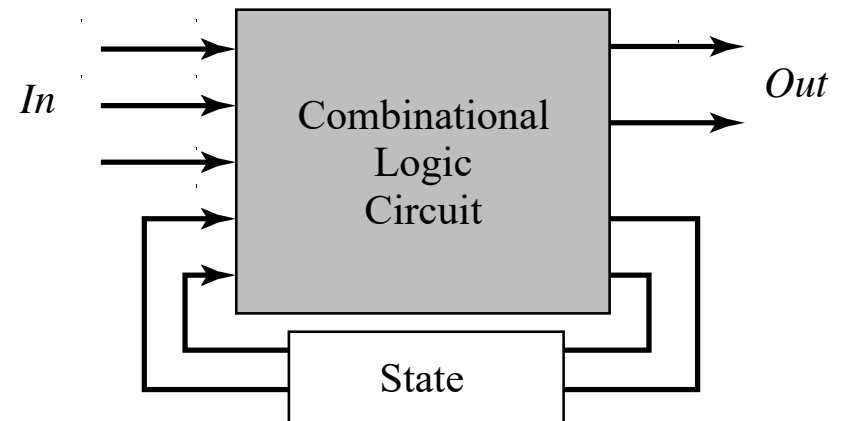
Designing Combinational Logic Circuits

Combinational vs. Sequential Logic



Combinational

$$\text{Output} = f(\text{In})$$



Sequential

$$\text{Output} = f(\text{In}, \text{Previous In})$$

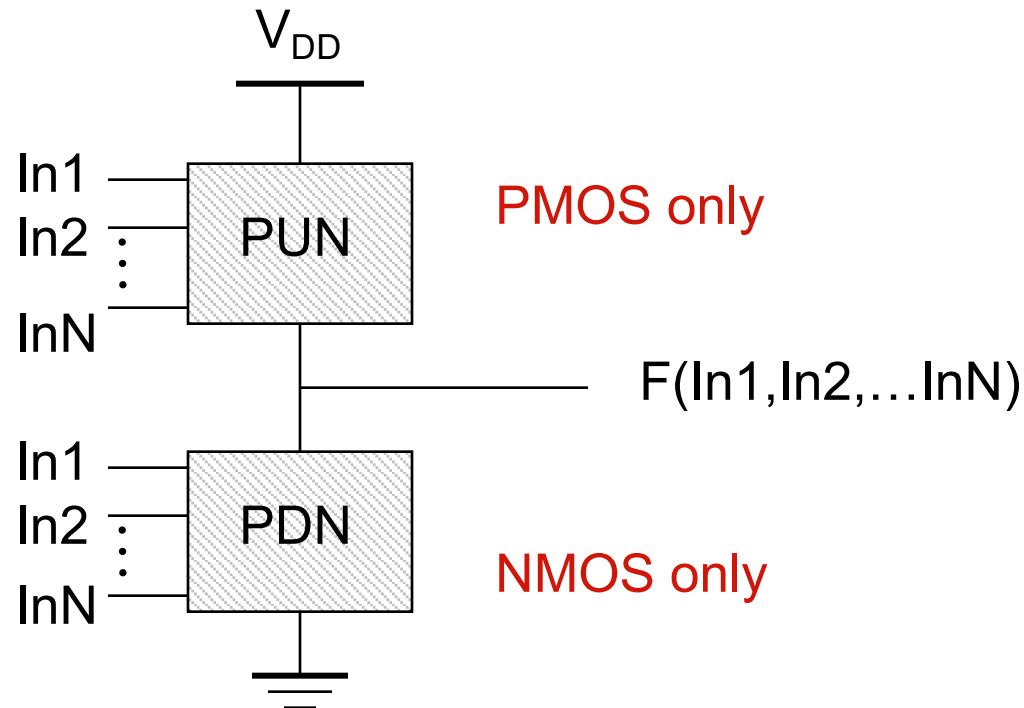
Static CMOS Circuit

At every point in time (except during the switching transients) each **gate output is connected to either V_{DD} or V_{ss}** via a low-resistive path.

The outputs of the gates **assume at all times the value of the Boolean function**, implemented by the circuit (ignoring, once again, the transient effects during switching periods).

This is in contrast to the *dynamic* circuit class, which relies on temporary storage of signal values on the capacitance of high impedance circuit nodes.

Static Complementary CMOS



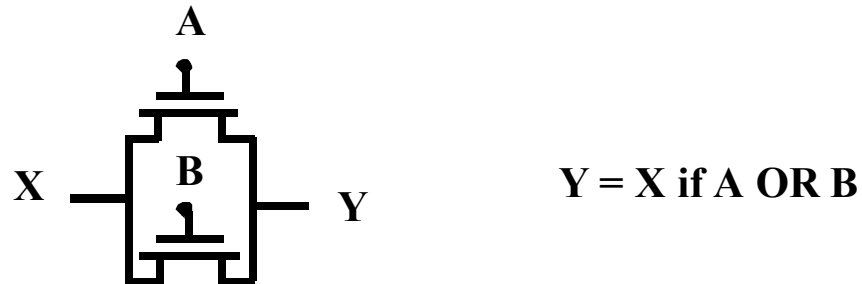
PUN and PDN are **dual** logic networks

One and only one of the networks (PUN or PDN) is conducting in steady state

NMOS Transistors in Series/Parallel Connection

Transistors can be thought as a switch controlled by its gate signal

NMOS switch closes when switch control input is high

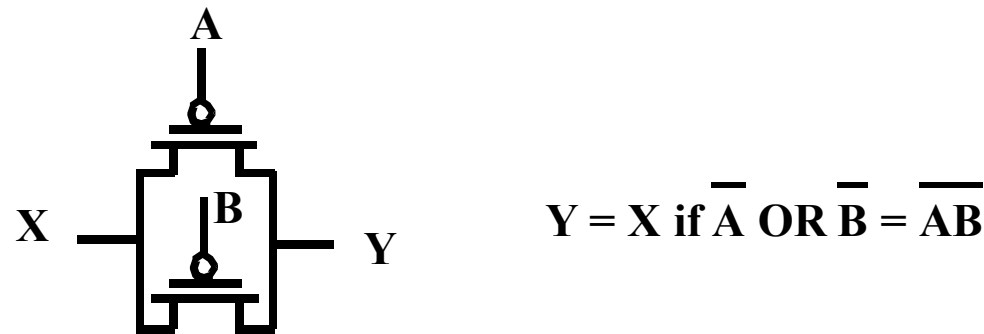
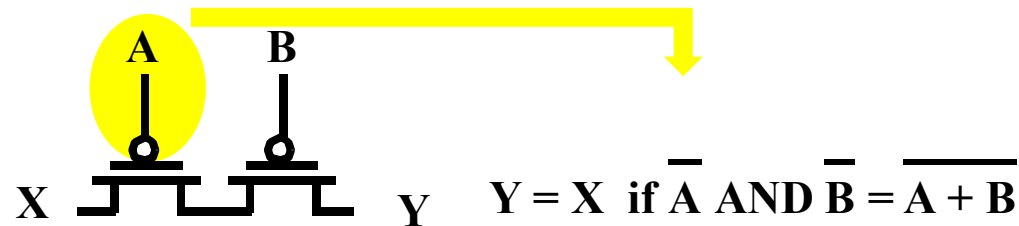


NMOS Transistors pass a “strong” 0 but a “weak” 1

CMOS is inverting logic

PMOS Transistors in Series/Parallel Connection

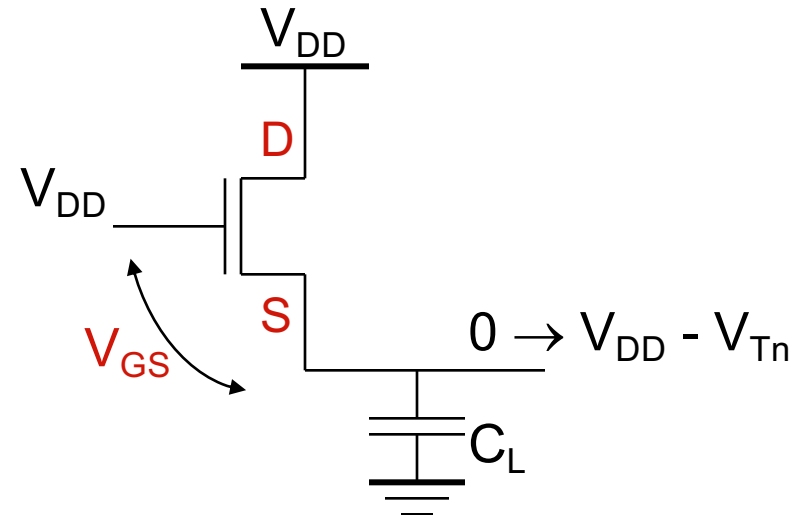
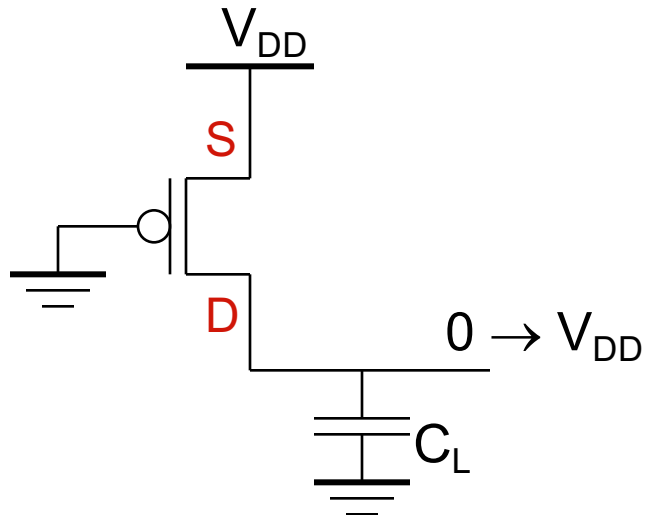
PMOS switch closes when switch control input is low



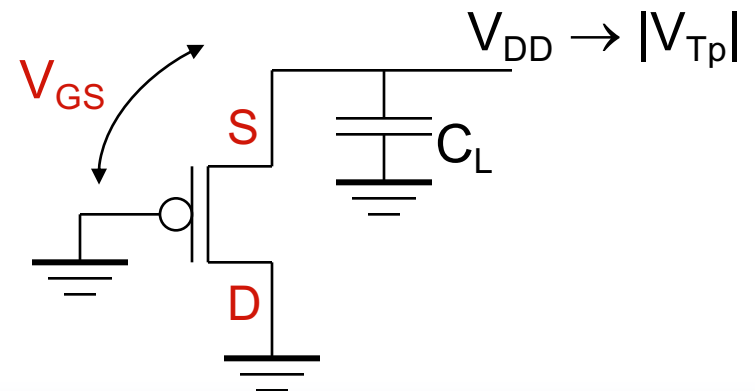
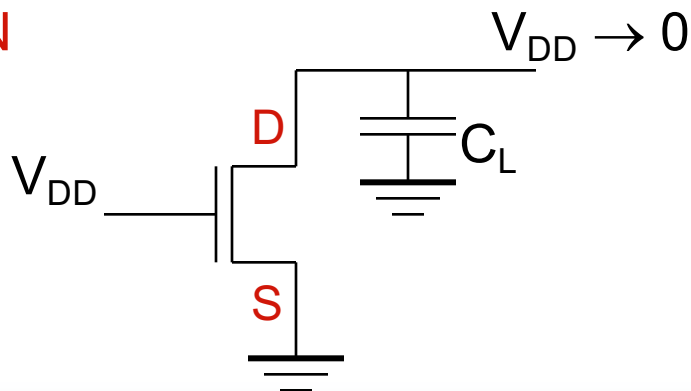
PMOS Transistors pass a “strong” 1 but a “weak” 0

Threshold Drops

PUN



PDN



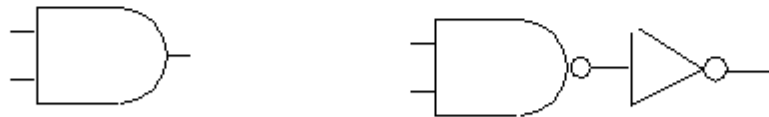
Complementary CMOS Logic Style

- PUP is the DUAL of PDN
(can be shown using DeMorgan's Theorem's)

$$\overline{A + B} = \bar{A}\bar{B}$$

$$\overline{AB} = \bar{A} + \bar{B}$$

- The complementary gate is inverting

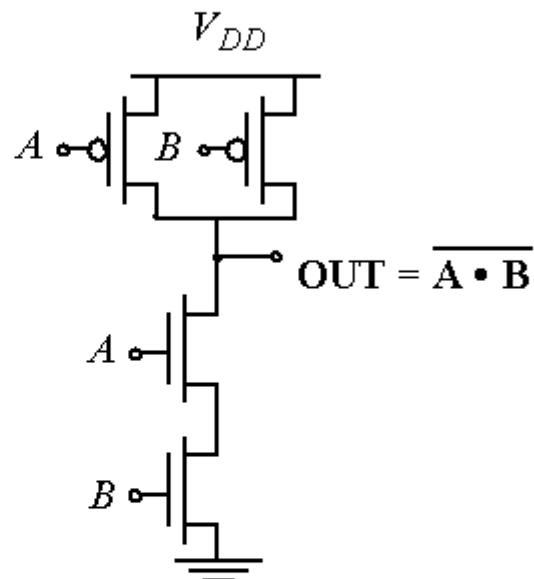


$$\text{AND} = \text{NAND} + \text{INV}$$

Example Gate: NAND

A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of a 2 input NAND gate



PDN: $G = A B \Rightarrow$ Conduction to GND

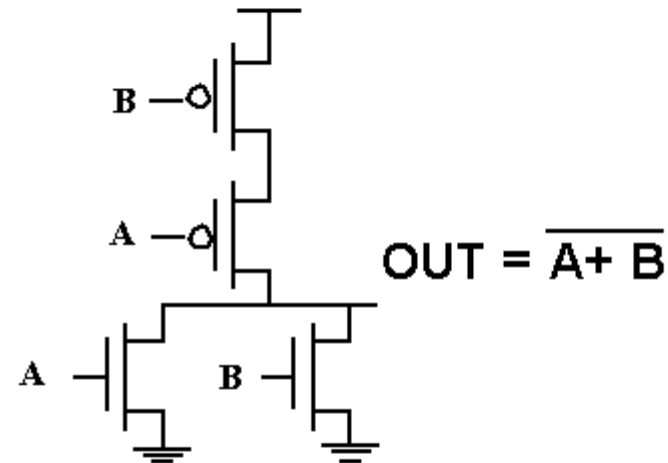
PUN: $F = \overline{A} + \overline{B} = \overline{AB} \Rightarrow$ Conduction to V_{DD}

$$\overline{G(In_1, In_2, In_3, \dots)} \equiv F(\overline{In_1}, \overline{In_2}, \overline{In_3}, \dots)$$

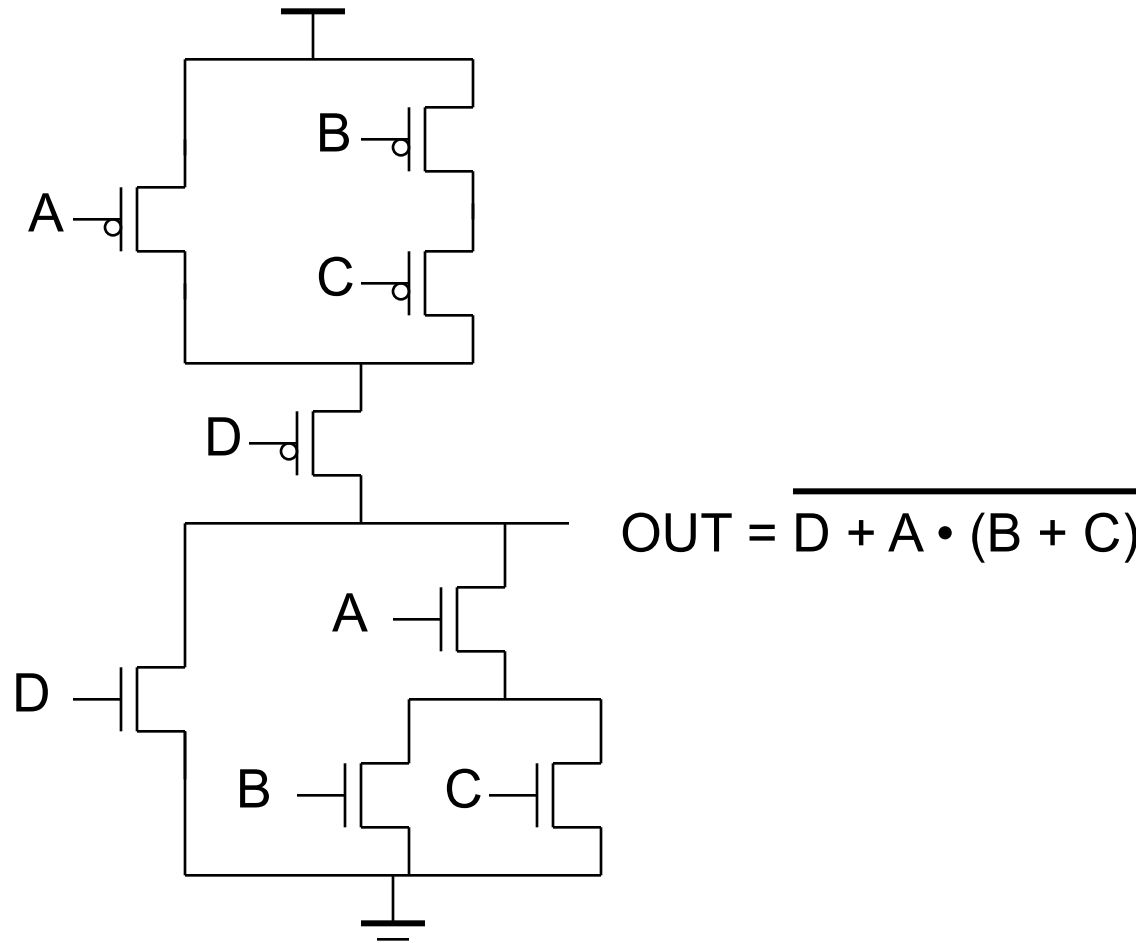
Example Gate: NOR

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

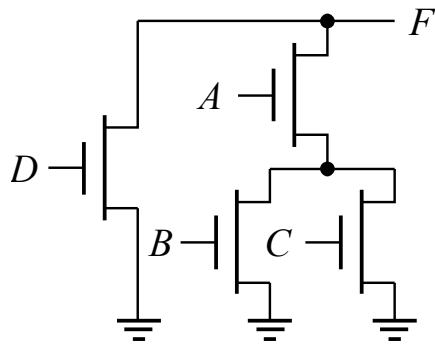
Truth Table of a 2 input NOR gate



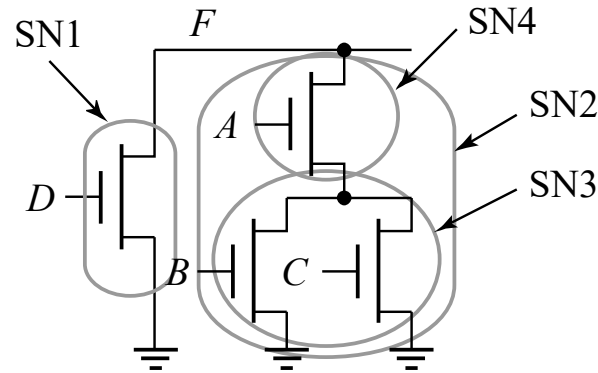
Complex CMOS Gate



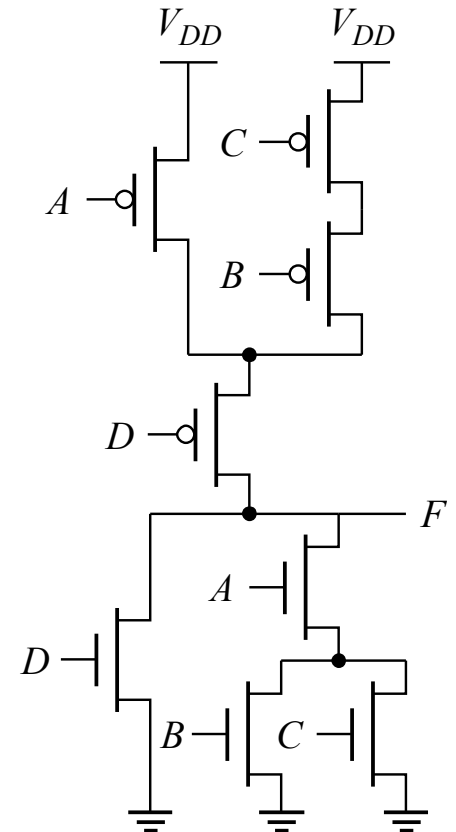
Constructing a Complex Gate



(a) pull-down network



sub-nets



(c) complete gate

Cell Design

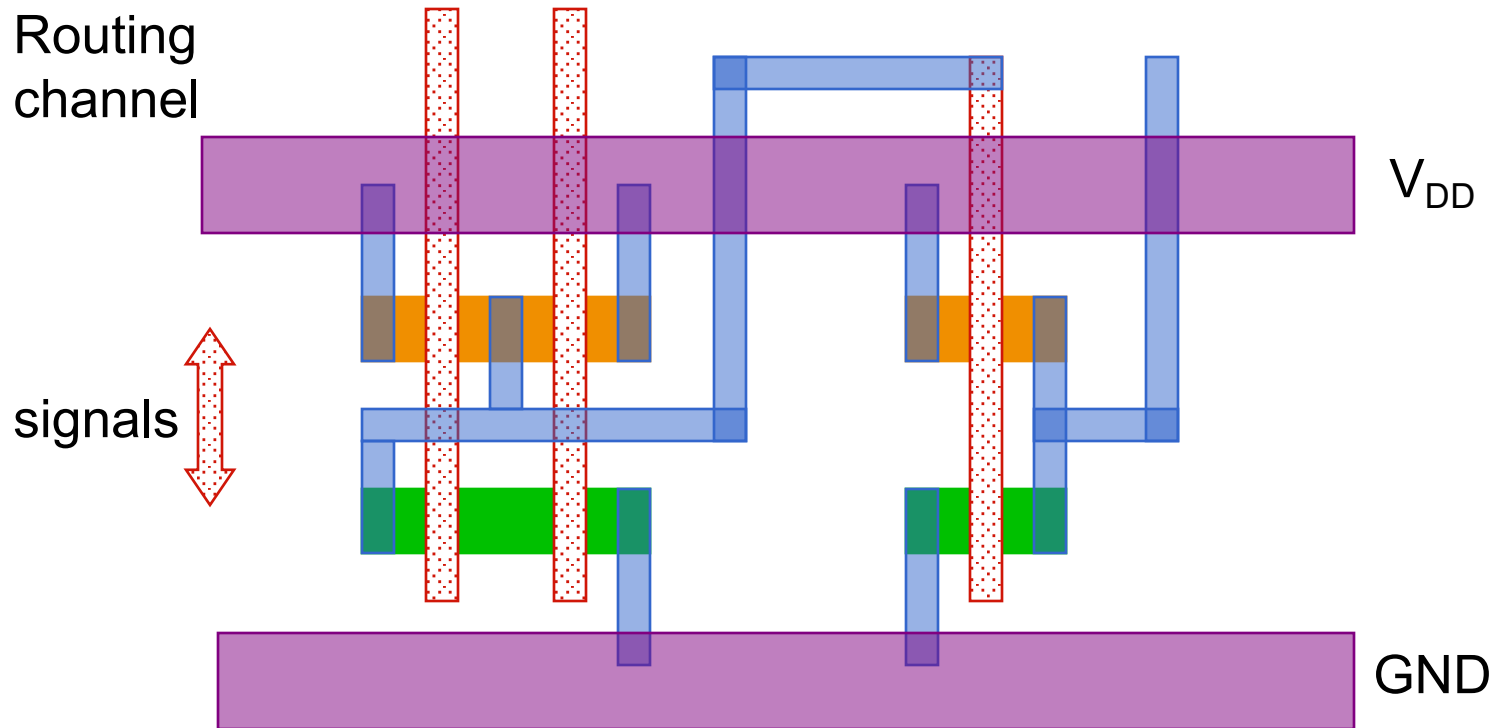
□ Standard Cells

- General purpose logic
- Can be synthesized
- Same height, varying width

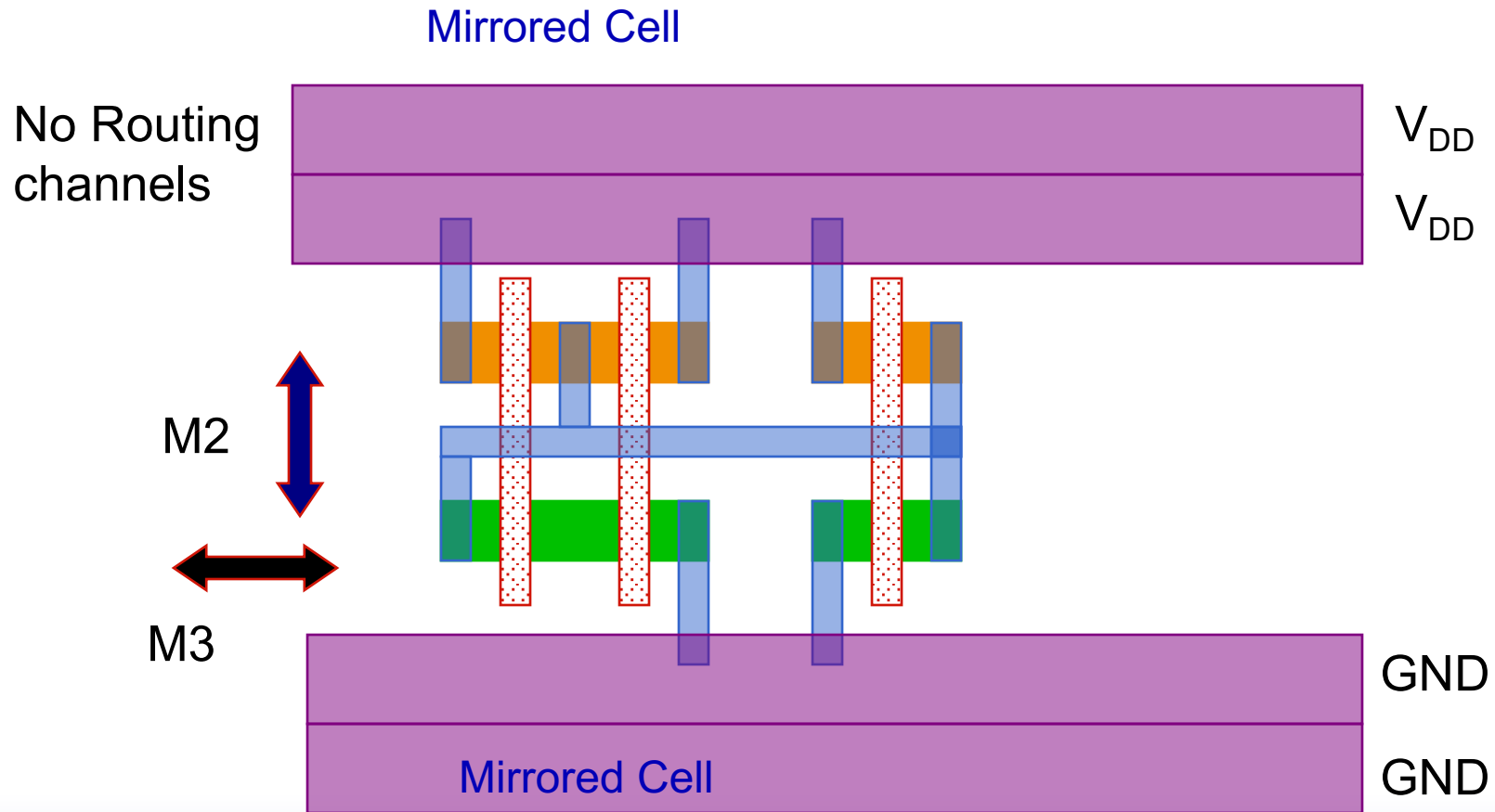
□ Datapath Cells

- For regular, structured designs (arithmetic)
- Includes some wiring in the cell
- Fixed height and width

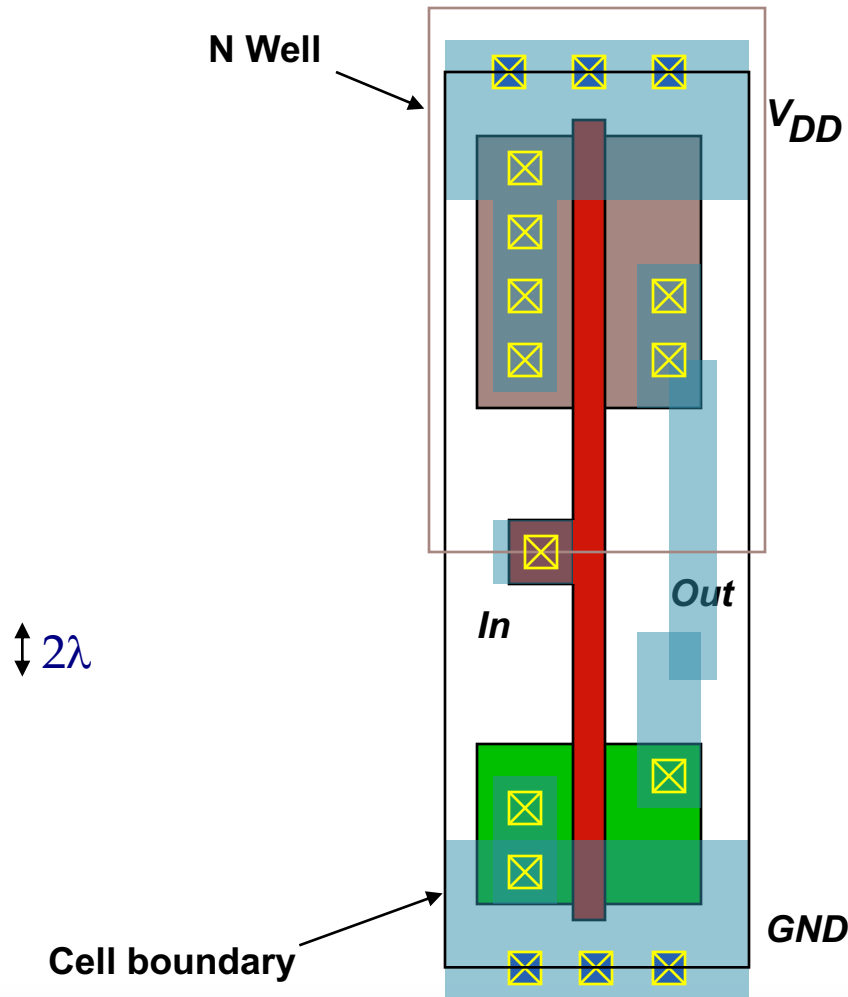
Standard Cell Layout Methodology – 1980s



Standard Cell Layout Methodology – 1990s



Standard Cells

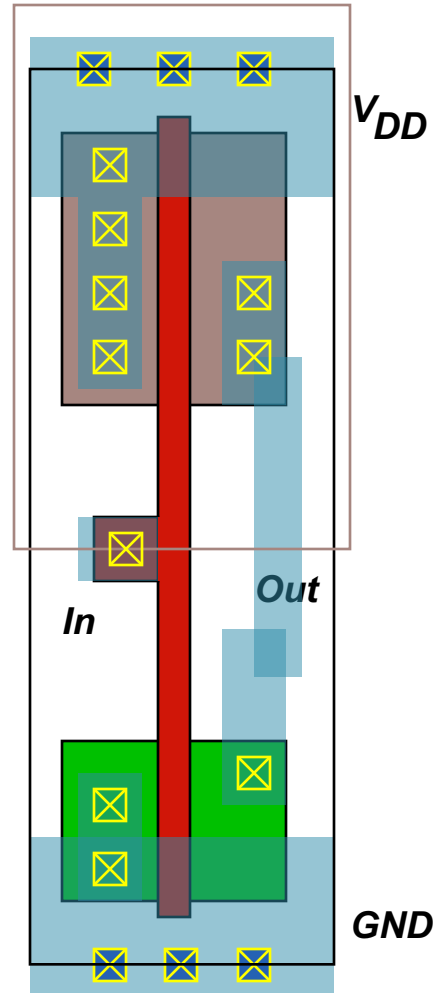
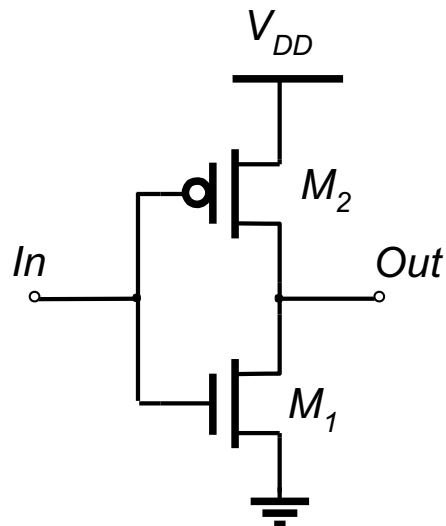


Cell height 12 metal tracks
Metal track is approx. $3\lambda + 3\lambda$
Pitch =
repetitive distance between objects
Cell height is "12 pitch"

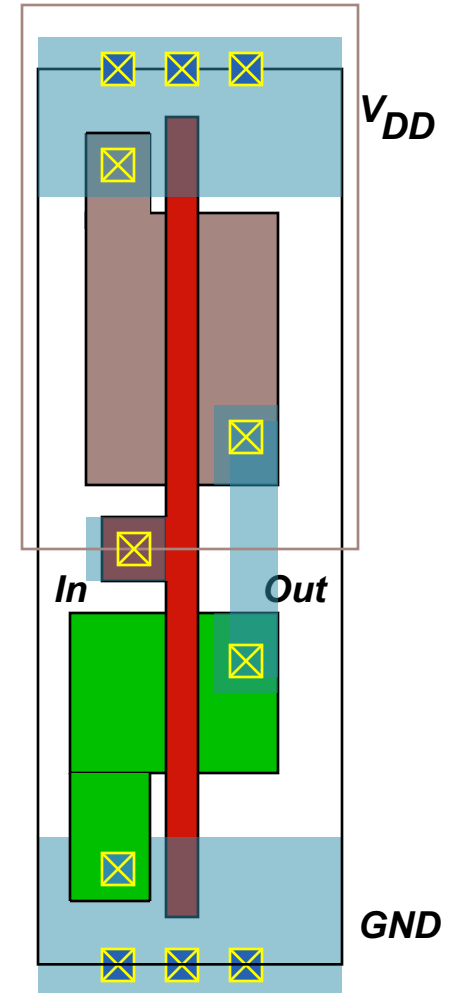
Rails $\sim 10\lambda$

Standard Cells

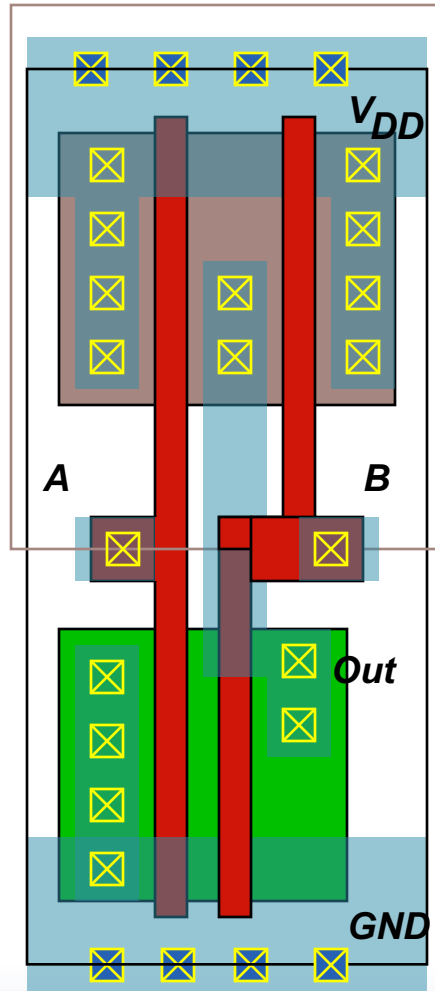
With minimal
diffusion
routing



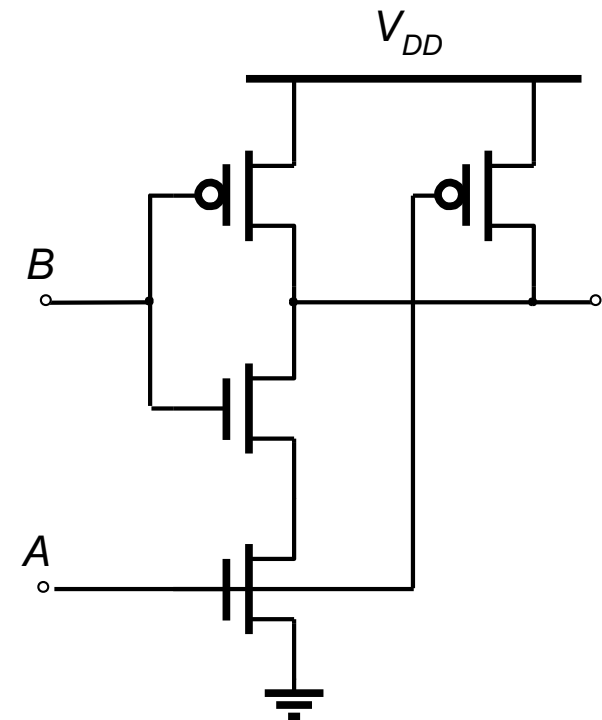
With silicided
diffusion



Standard Cells



2-input NAND gate

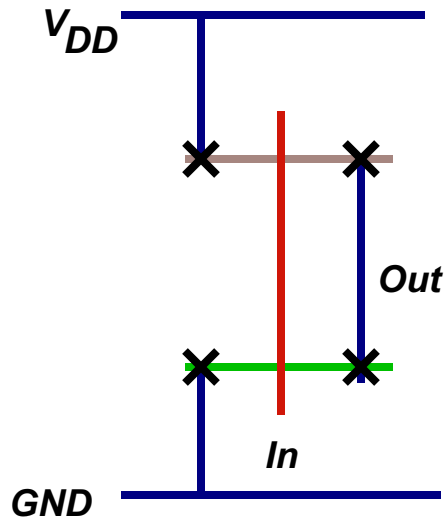


Stick Diagrams

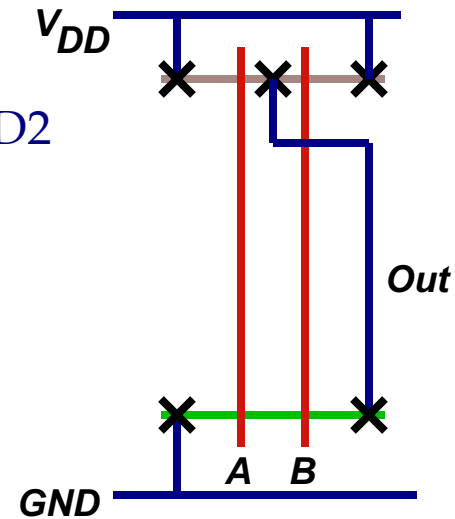
Contains no dimensions

Represents relative positions of transistors

Inverter

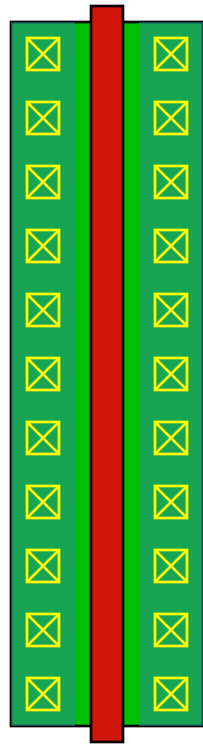


NAND2

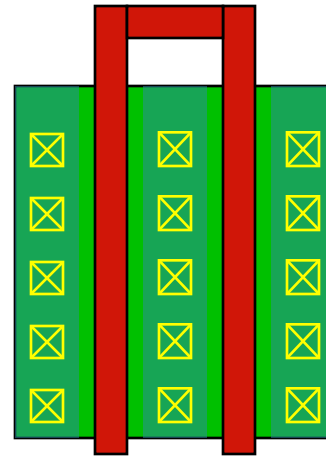


Multi-Fingered Transistors

One finger



Two fingers (folded)



Less diffusion capacitance

Properties of Complementary CMOS Gates

Snapshot

High noise margins:

V_{OH} and V_{OL} are at V_{DD} and GND , respectively.

No static power consumption:

There never exists a direct path between V_{DD} and V_{SS} (GND) in steady-state mode.

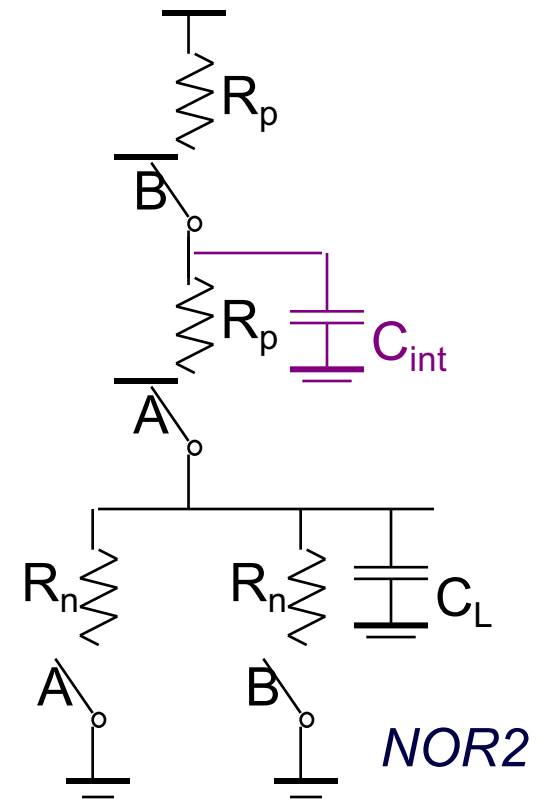
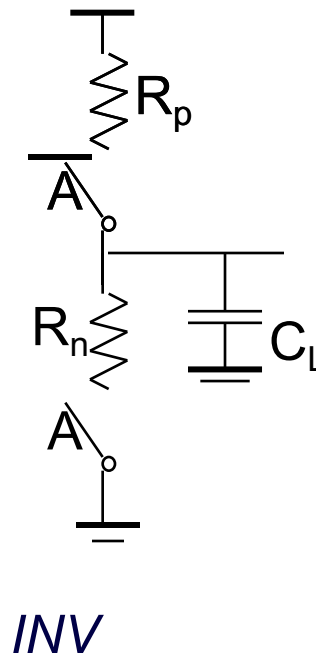
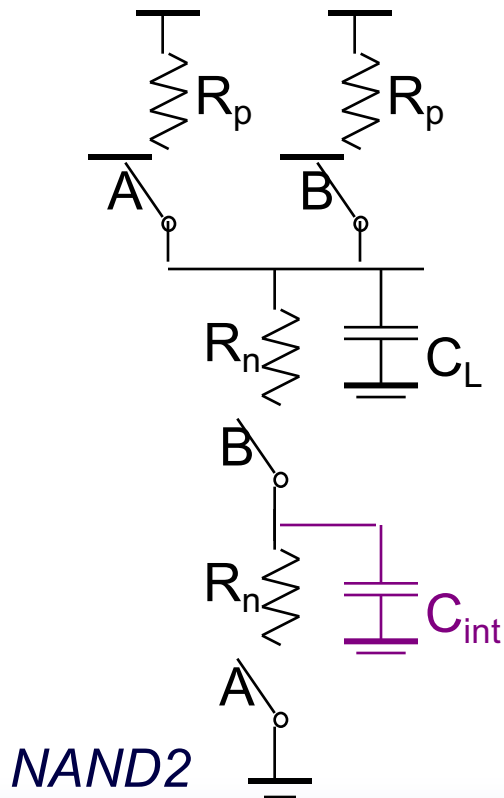
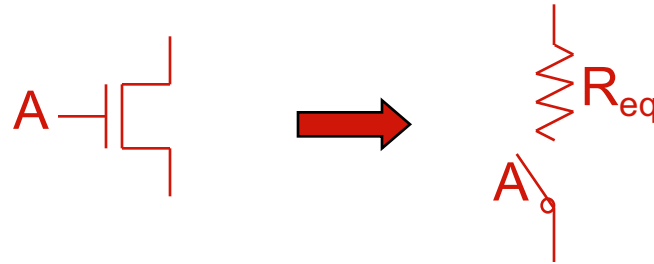
Comparable rise and fall times:

(under appropriate **sizing conditions**)

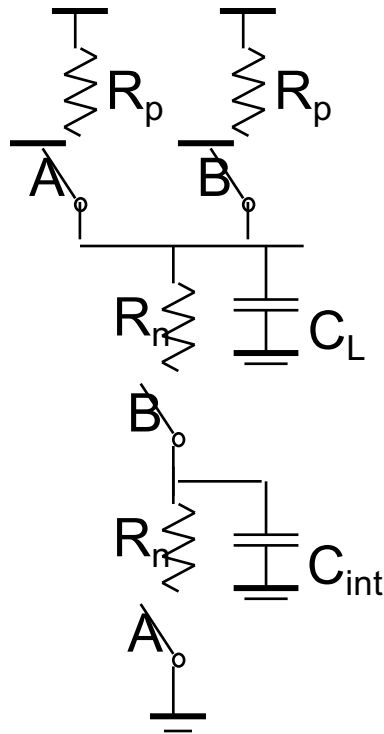
CMOS Properties

- ❑ Full rail-to-rail swing; **high noise margins**
- ❑ Logic levels not dependent upon the relative device sizes; **ratioless**
- ❑ Always a path to Vdd or Gnd in steady state; **low output impedance**
- ❑ Extremely **high input resistance**; nearly zero steady-state input current
- ❑ No direct path steady state between power and ground; **no static power dissipation**
- ❑ Propagation delay function of load capacitance and resistance of transistors

Switch Delay Model

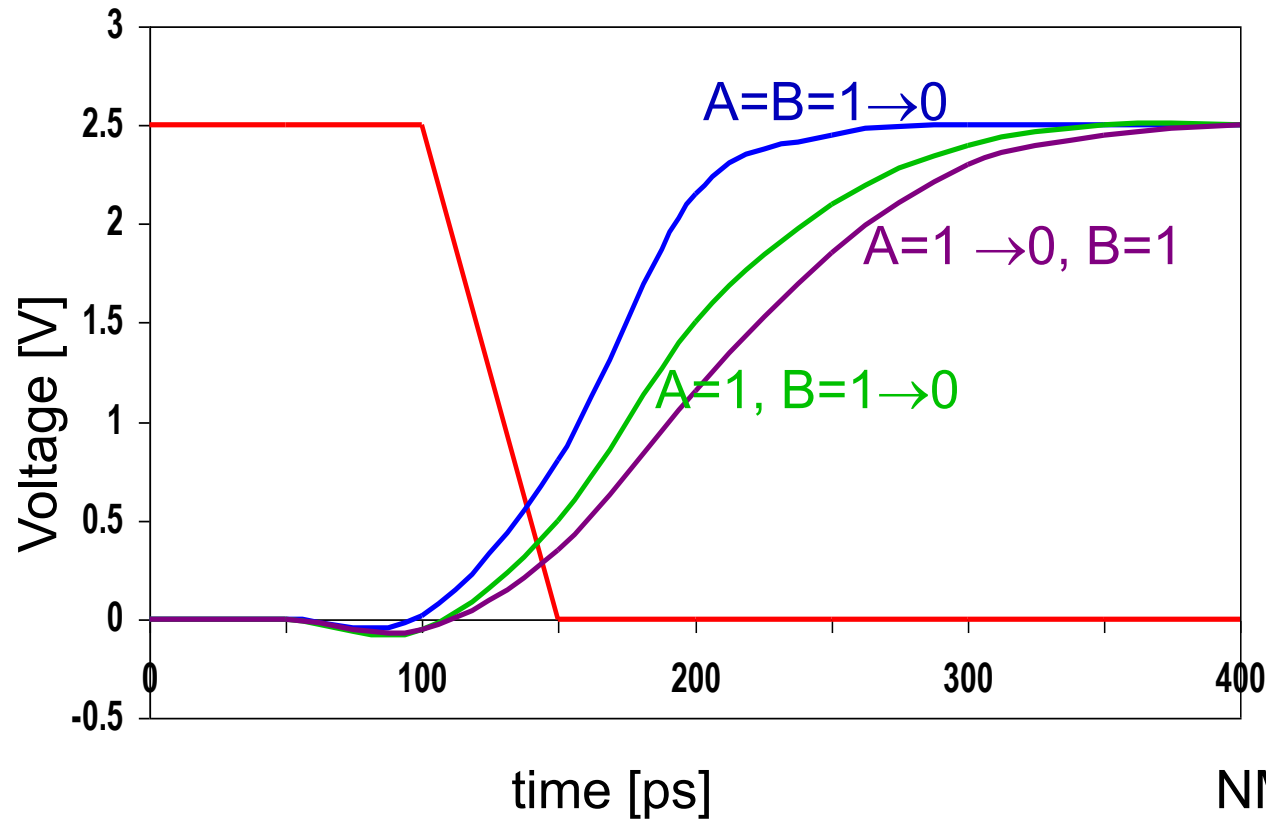


Input Pattern Effects on Delay



- Delay is dependent on the **pattern** of inputs
- Low to high transition
 - both inputs go low
 - delay is $0.69 R_p/2 C_L$
 - one input goes low
 - delay is $0.69 R_p C_L$
- High to low transition
 - both inputs go high
 - delay is $0.69 2R_n C_L$

Delay Dependence on Input Patterns



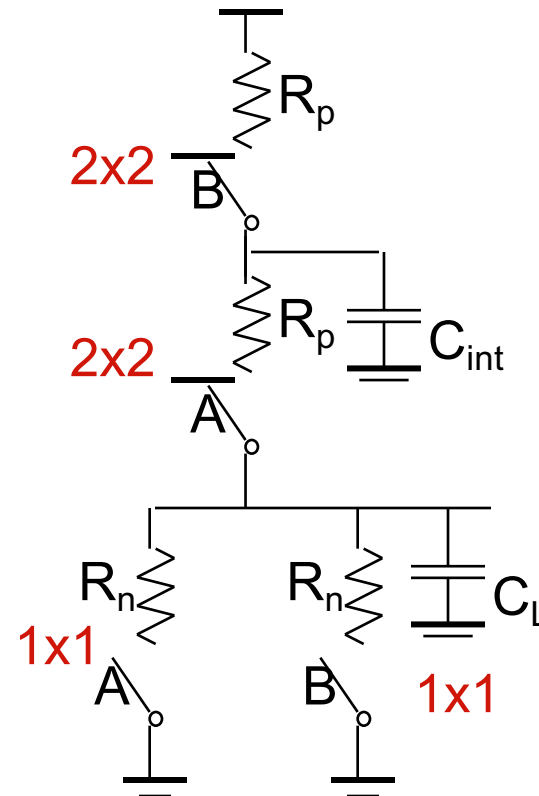
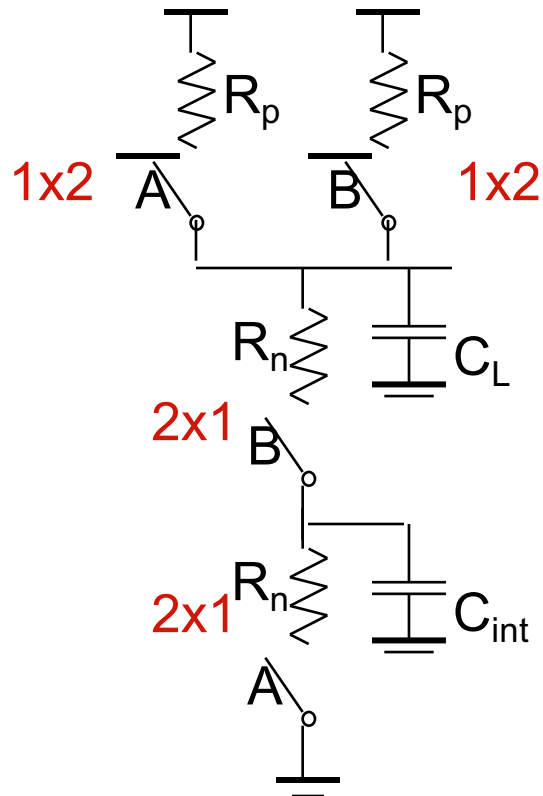
Estimates of delay can be fairly complex – have to consider internal node capacitances and the data patterns.

Input Data Pattern	Delay (psec)
$A=B=0 \rightarrow 1$	67
$A=1, B=0 \rightarrow 1$	64
$A=0 \rightarrow 1, B=1$	61
$A=B=1 \rightarrow 0$	45
$A=1, B=1 \rightarrow 0$	80
$A=1 \rightarrow 0, B=1$	81

NMOS = $0.5\mu\text{m}/0.25\mu\text{m}$
PMOS = $0.75\mu\text{m}/0.25\mu\text{m}$
 $C_L = 100\text{ fF}$

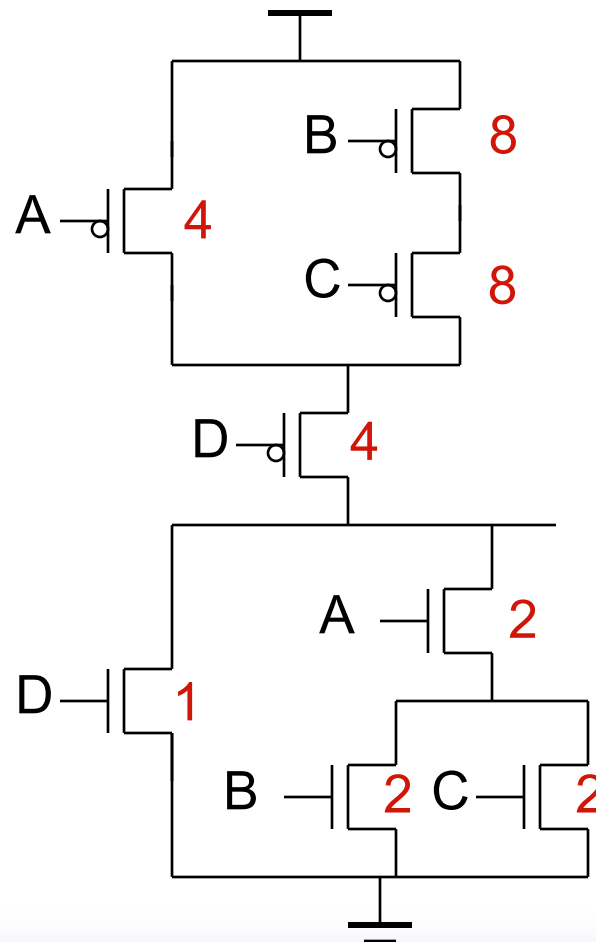
Transistor Sizing

Hp. $R_p = 2R_n$



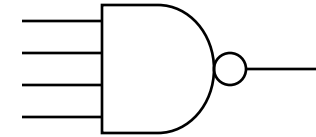
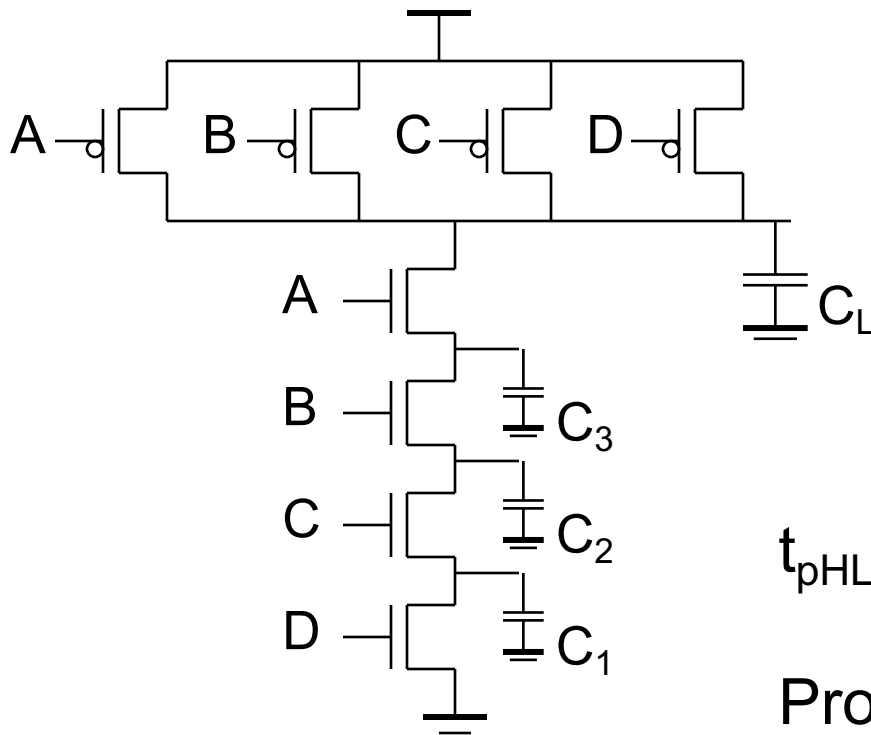
Transistor Sizing a Complex CMOS Gate

$$H_p. R_p = 2R_n$$



$$OUT = \overline{D + A \cdot (B + C)}$$

Fan-In Considerations

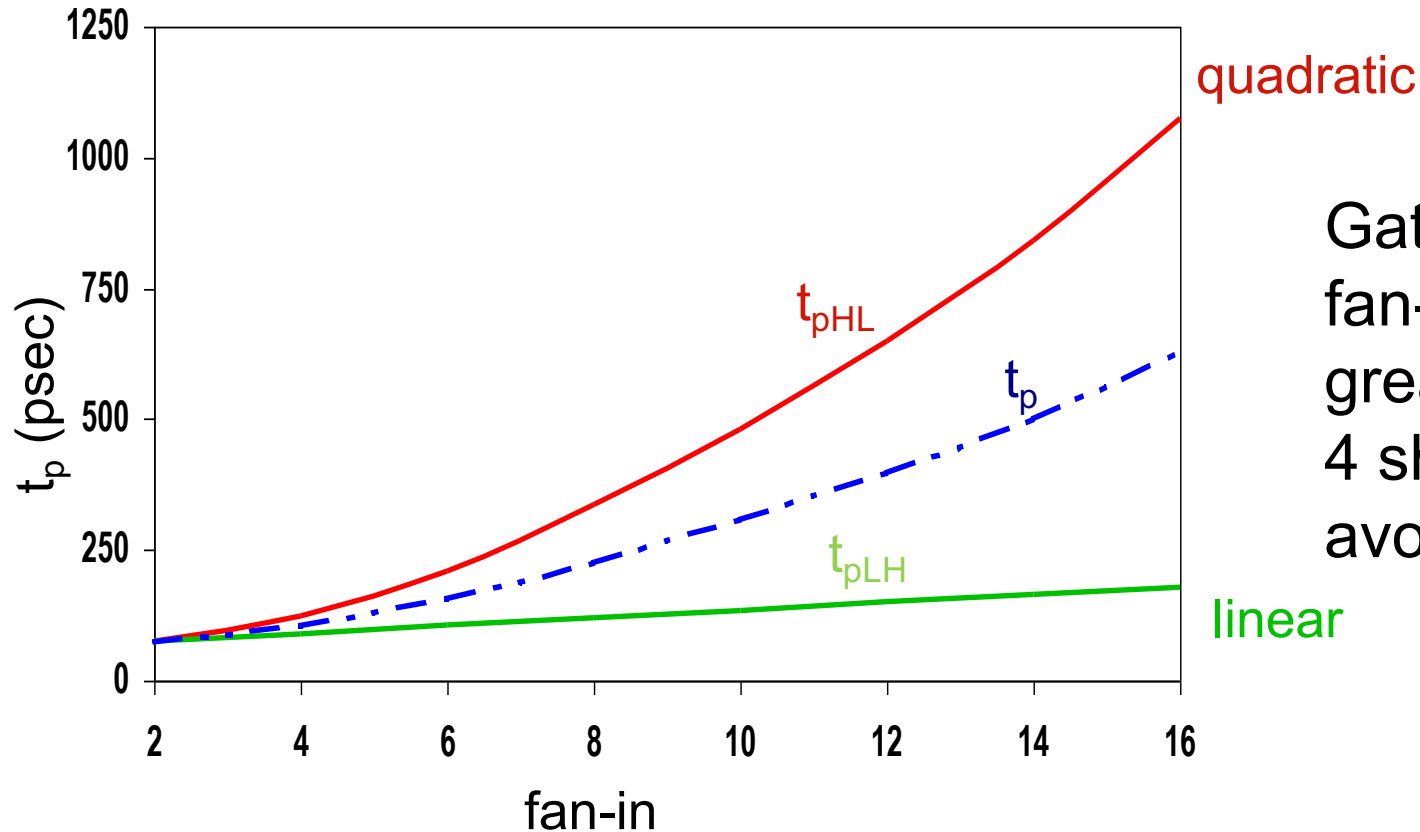


Distributed RC model
(Elmore delay)

$$t_{pHL} = 0.69 R_{eqn}(C_1 + 2C_2 + 3C_3 + 4C_L)$$

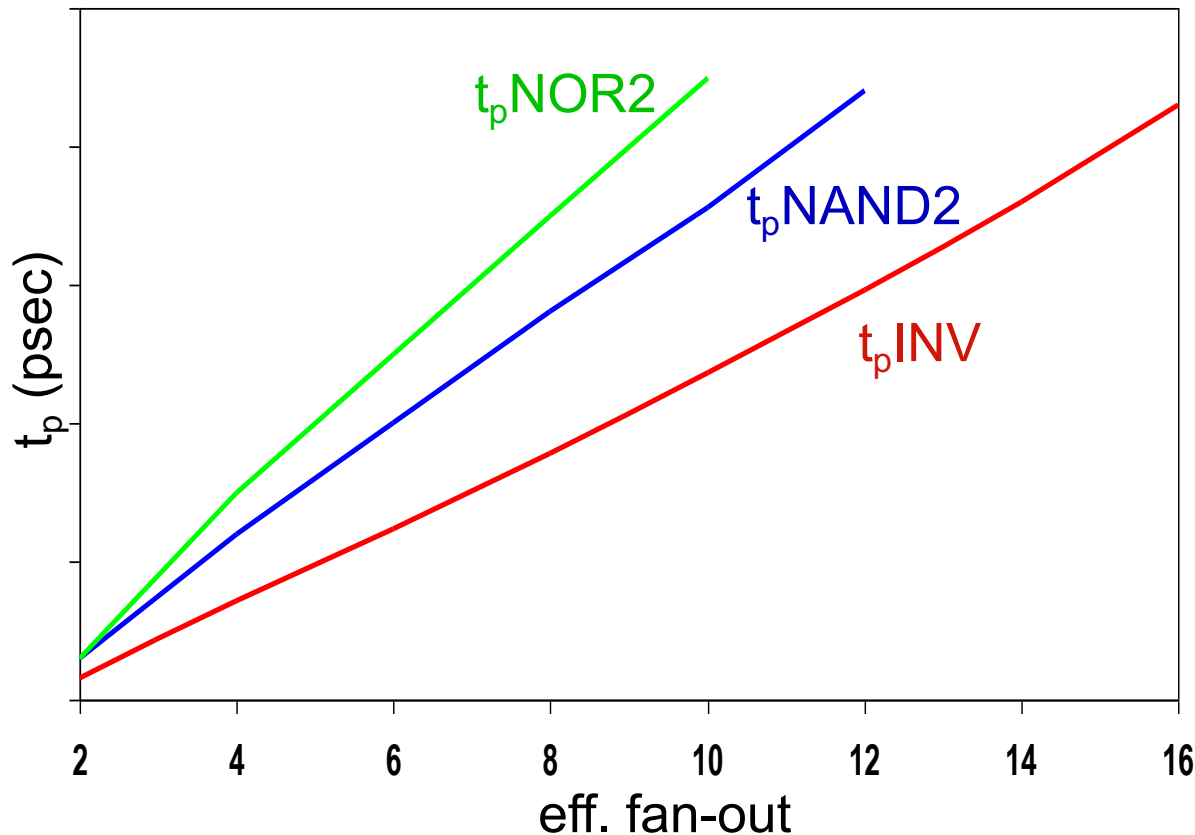
Propagation delay deteriorates rapidly as a function of fan-in – **quadratically** in the worst case.

t_p as a Function of Fan-In



Gates with a fan-in greater than 4 should be avoided.

t_p as a Function of Fan-Out



All gates have the same drive current.

Slope is a function of “driving strength”

t_p as a Function of Fan-In and Fan-Out

- Fan-in: **quadratic** due to increasing resistance and capacitance
- Fan-out: each additional fan-out gate adds **two** gate capacitances to C_L

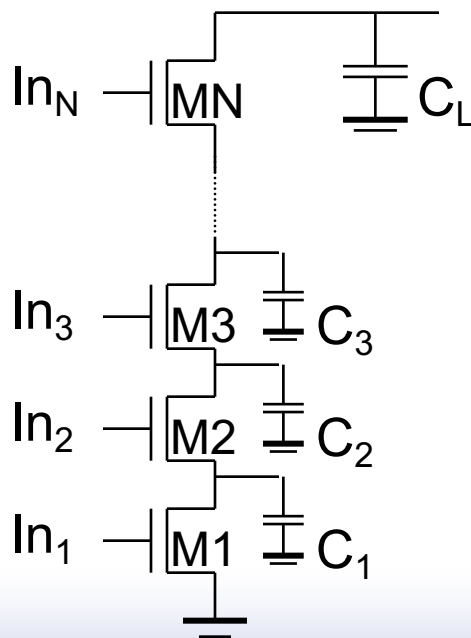
$$t_p = a_1 FI + a_2 FI^2 + a_3 FO$$

Fast Complex Gates: Design Technique 1

□ Transistor sizing

- as long as fan-out capacitance dominates

□ Progressive sizing



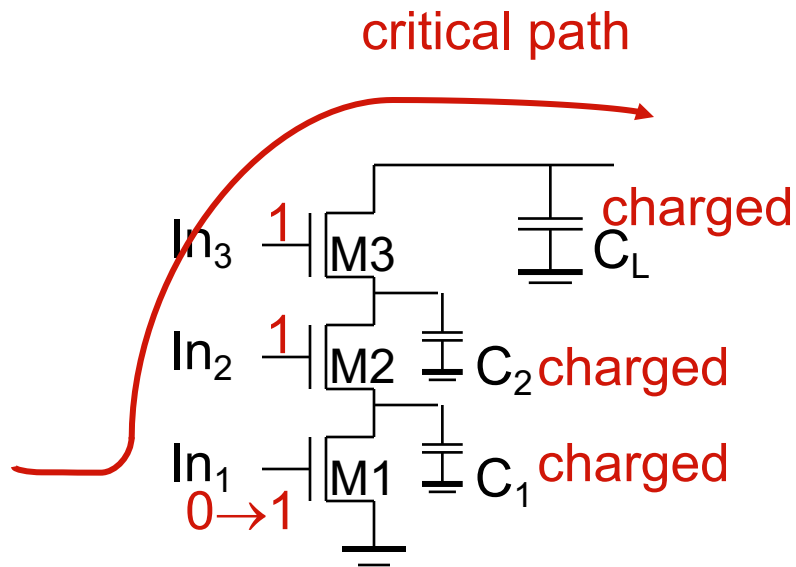
Distributed RC line

$M1 > M2 > M3 > \dots > MN$
(the fet closest to the output is the smallest)

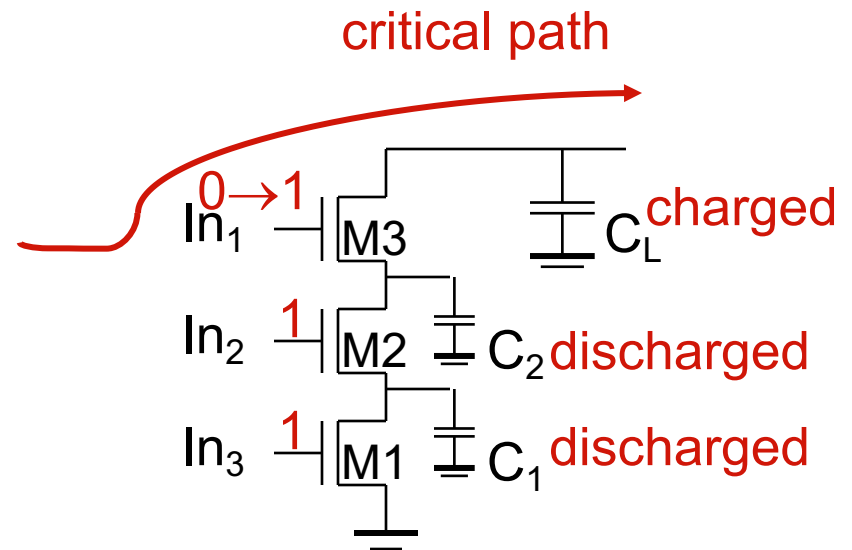
Can reduce delay by more than 20%; decreasing gains as technology shrinks

Fast Complex Gates: Design Technique 2

□ Transistor ordering



delay determined by time to
discharge C_L , C_1 and C_2

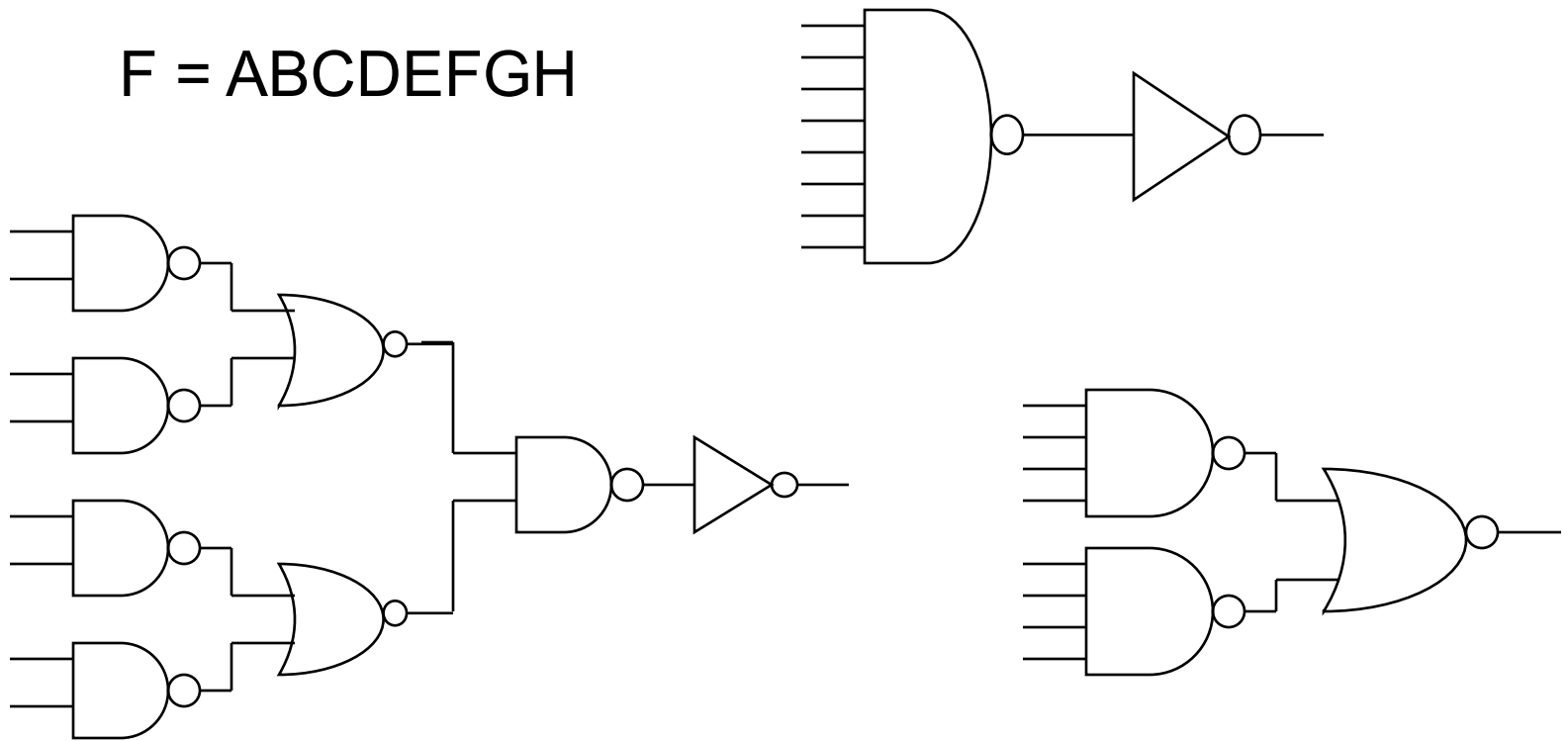


delay determined by time to
discharge C_L

Fast Complex Gates: Design Technique 3

□ Alternative logic structures

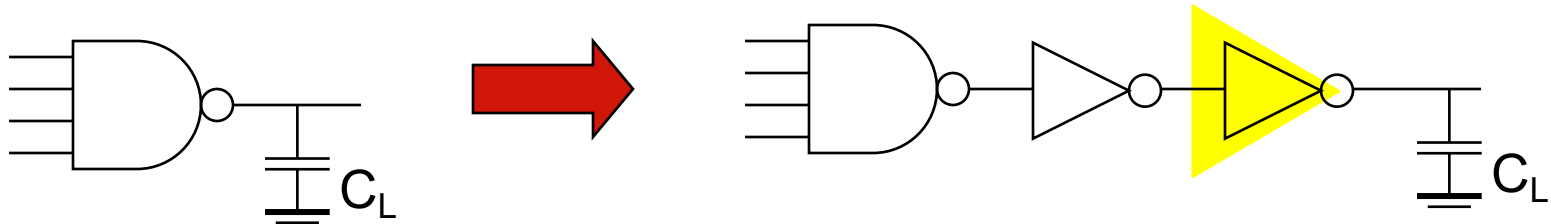
$$F = ABCDEFGH$$



Reduced fan-in -> deeper logic depth

Fast Complex Gates: Design Technique 4

- Isolating fan-in from fan-out using buffer insertion



Fast Complex Gates:

Design Technique 5

- Reducing the voltage swing

$$t_{pHL} = 0.69 (3/4 (C_L V_{DD}) / I_{DSATn})$$

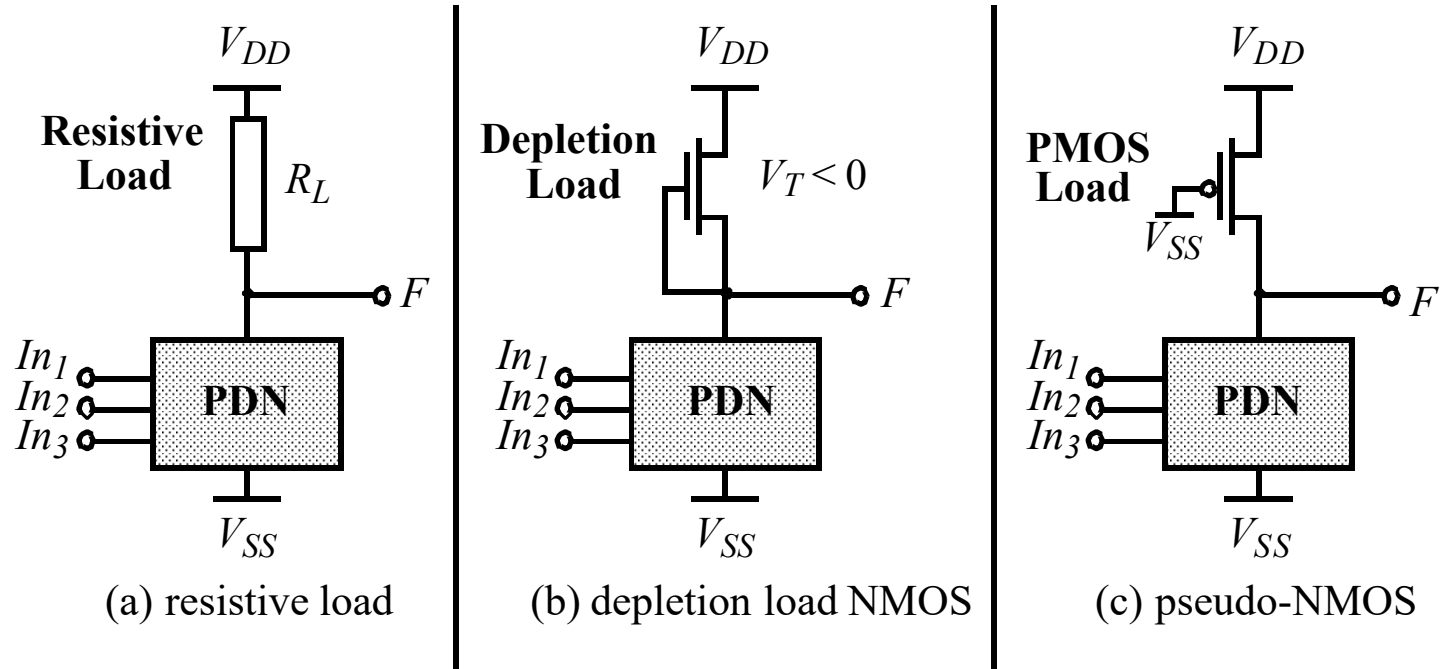
$$= 0.69 (3/4 (C_L V_{swing}) / I_{DSATn})$$

- linear reduction in delay
- also reduces power consumption
- But the following gate is much slower!
- Or requires use of “sense amplifiers” on the receiving end to restore the signal level (memory design)



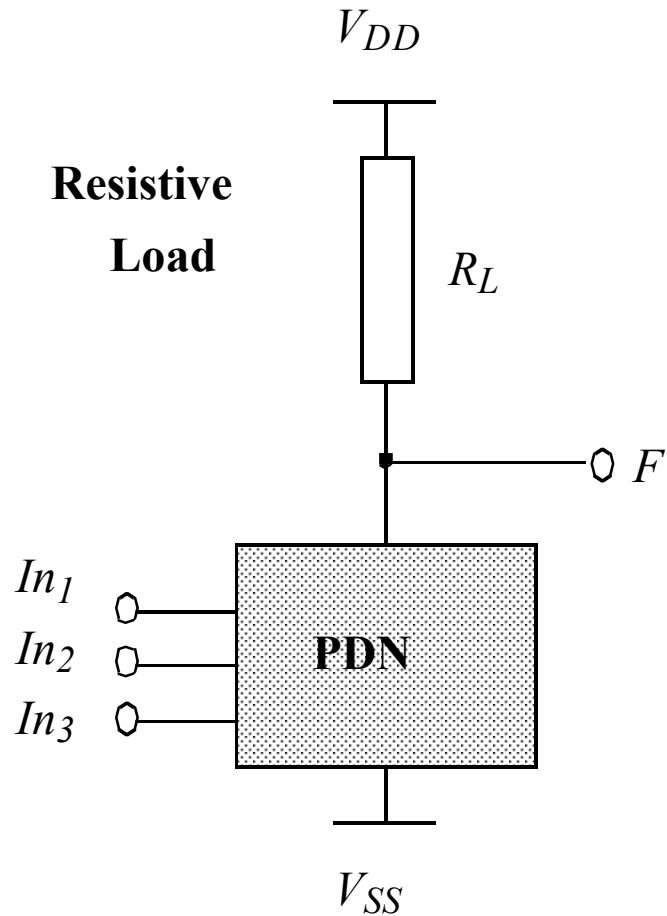
Ratioed Logic

Ratioed Logic



Goal: to reduce the number of devices over complementary CMOS

Ratioed Logic



- **N transistors + Load**

- **$V_{OH} = V_{DD}$**

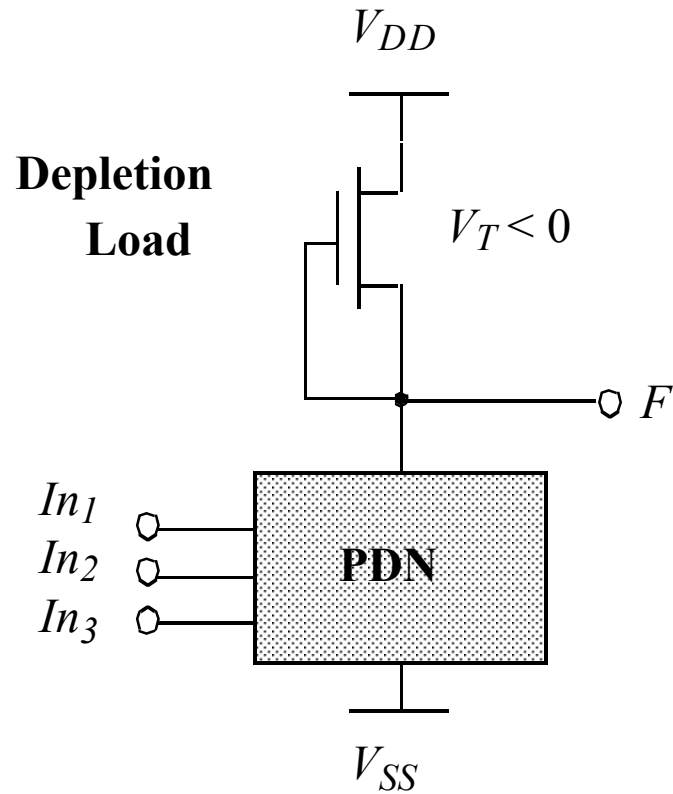
- **$V_{OL} = \frac{R_{PN}}{R_{PN} + R_L}$**

- **Assymetrical response**

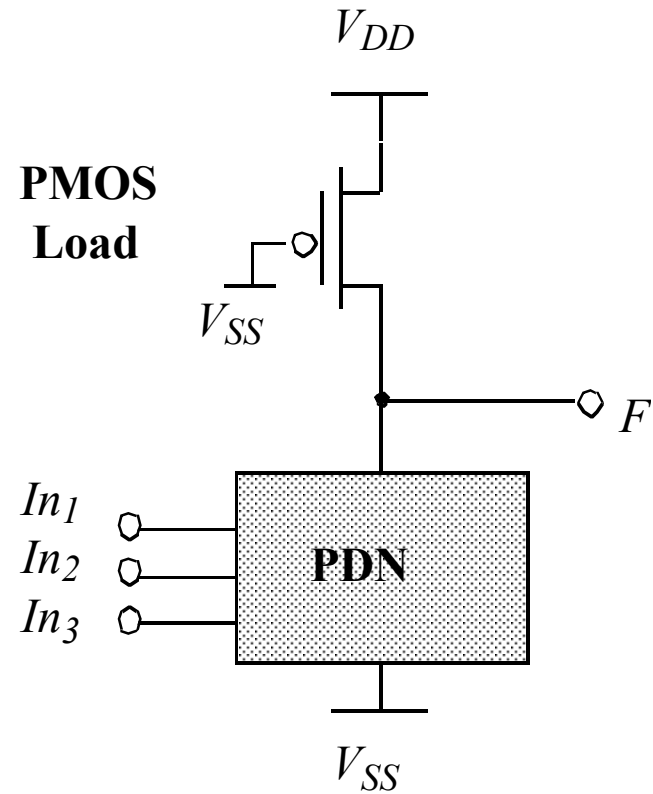
- **Static power consumption**

- **$t_{pL} = 0.69 R_L C_L$**

Active Loads

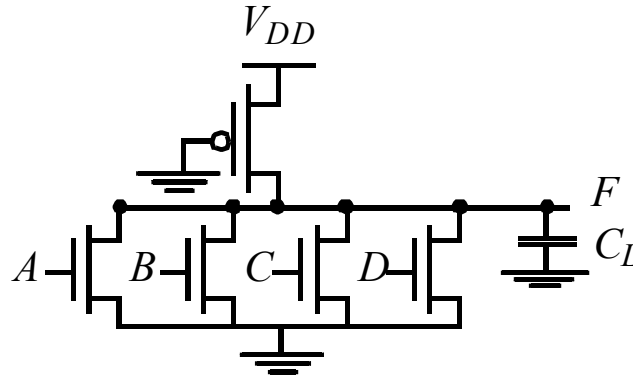


depletion load NMOS



pseudo-NMOS

Pseudo-NMOS



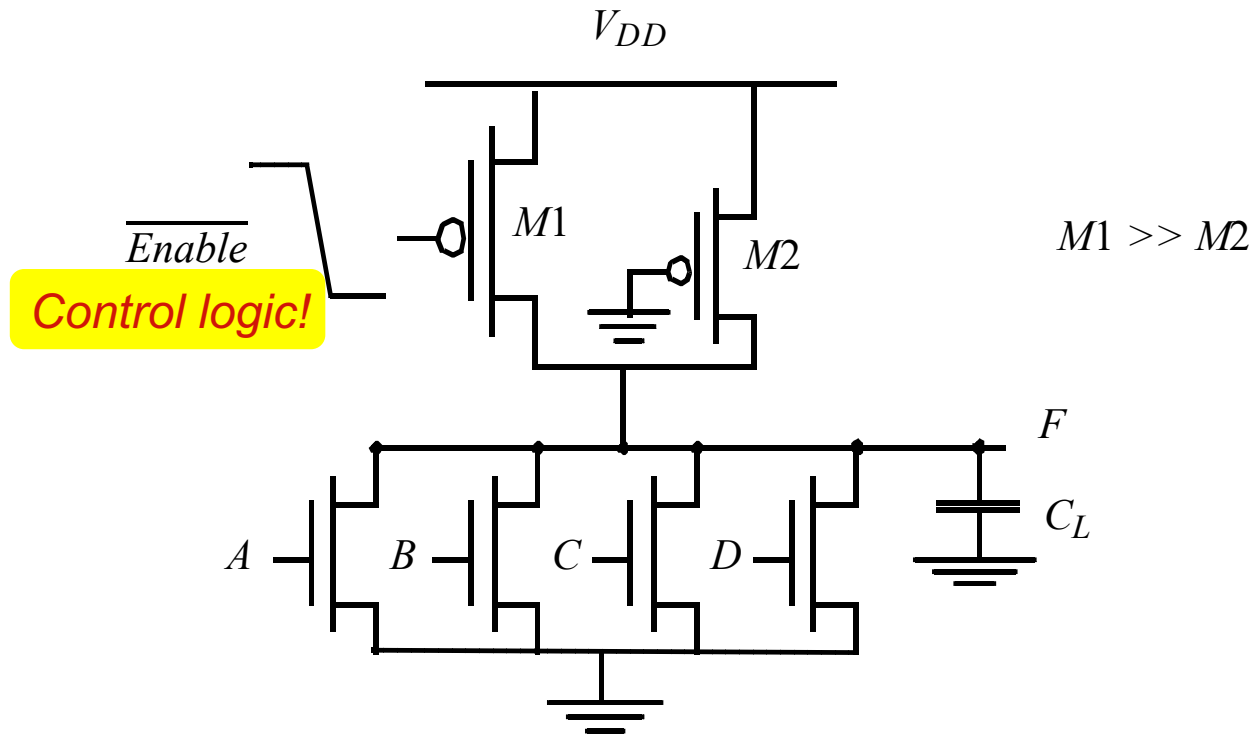
$$V_{OH} = V_{DD} \text{ (similar to complementary CMOS)}$$

$$k_n \left((V_{DD} - V_{Tn}) V_{OL} - \frac{V_{OL}^2}{2} \right) = \frac{k_p}{2} (V_{DD} - |V_{Tp}|)^2$$

$$V_{OL} = (V_{DD} - V_T) \left[1 - \sqrt{1 - \frac{k_p}{k_n}} \right] \text{ (assuming that } V_T = V_{Tn} = |V_{Tp}| \text{)}$$

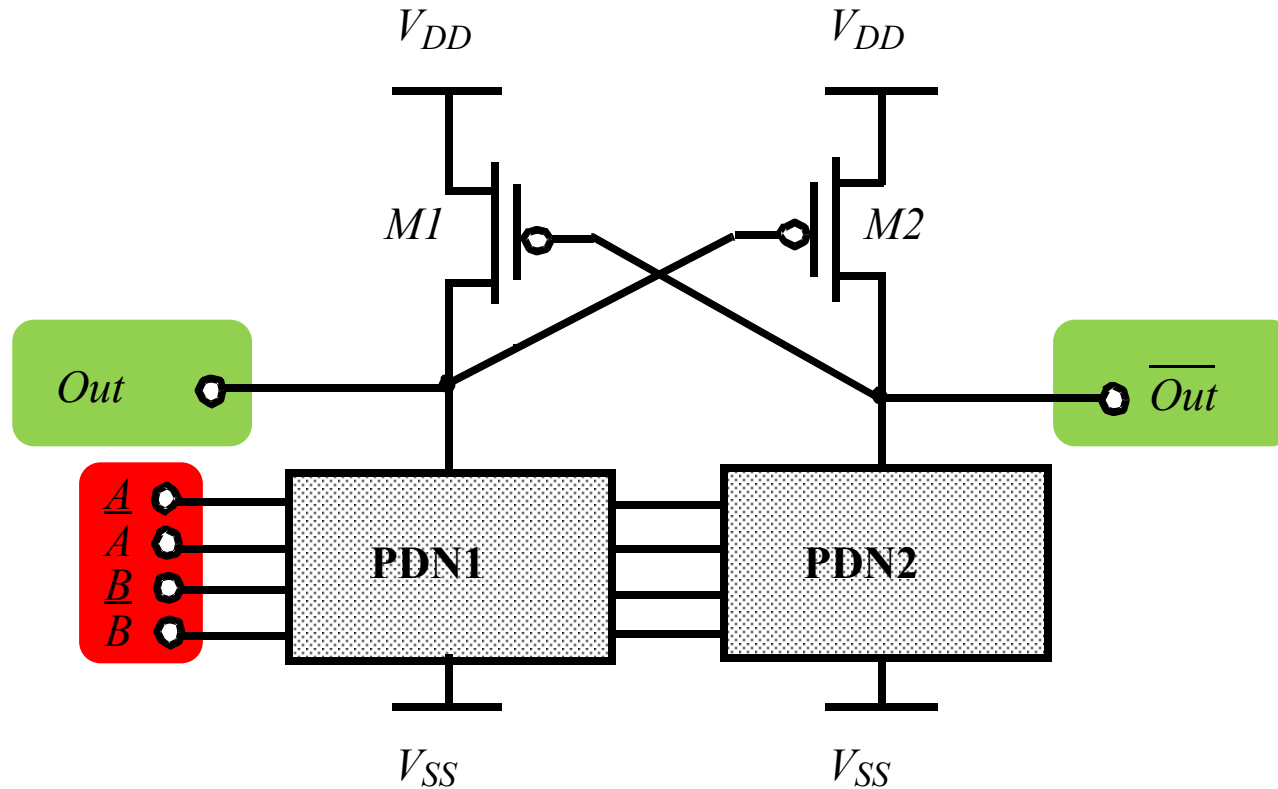
SMALLER AREA & LOAD BUT STATIC POWER DISSIPATION!!!

Improved Loads



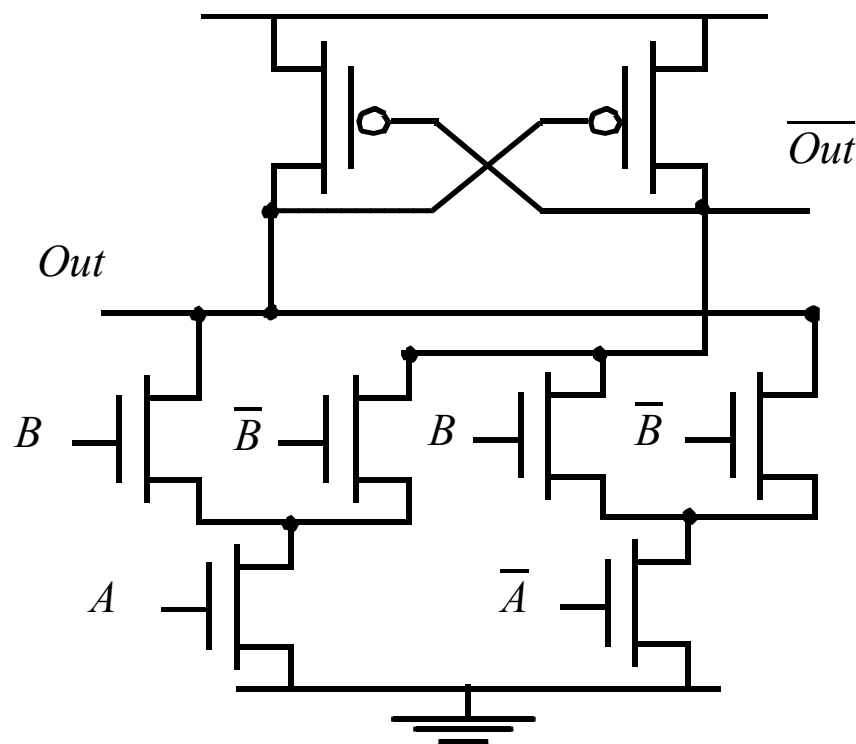
Adaptive Load

Improved Loads (2)



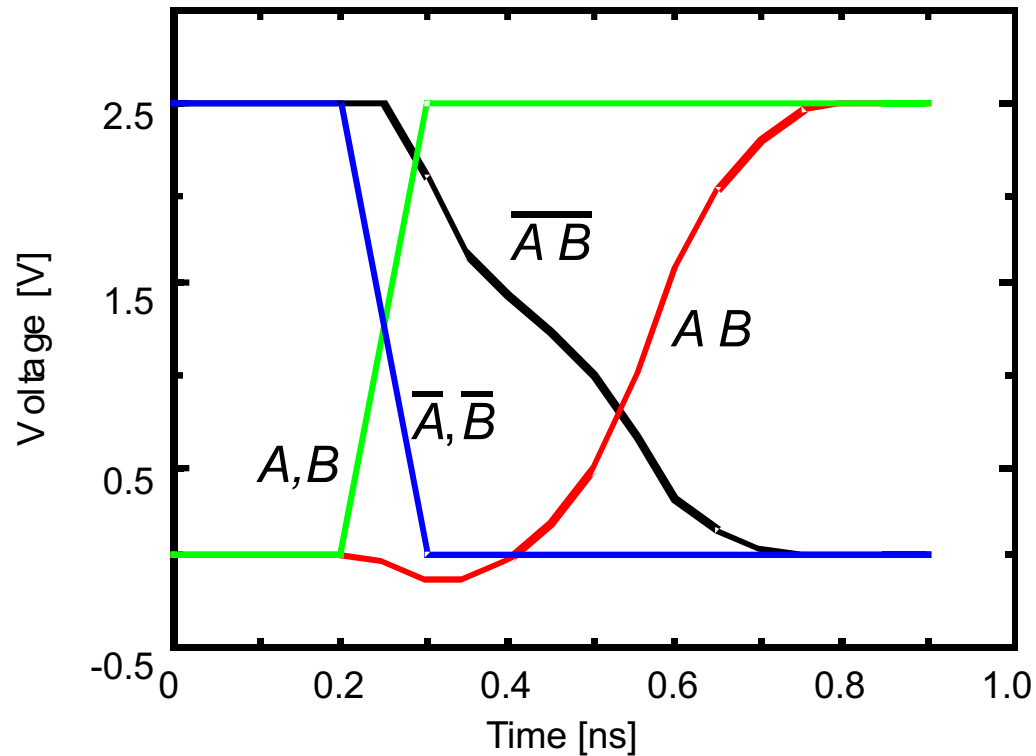
Differential Cascode Voltage Switch Logic (DCVSL)

DCVSL Example



XOR-NXOR gate

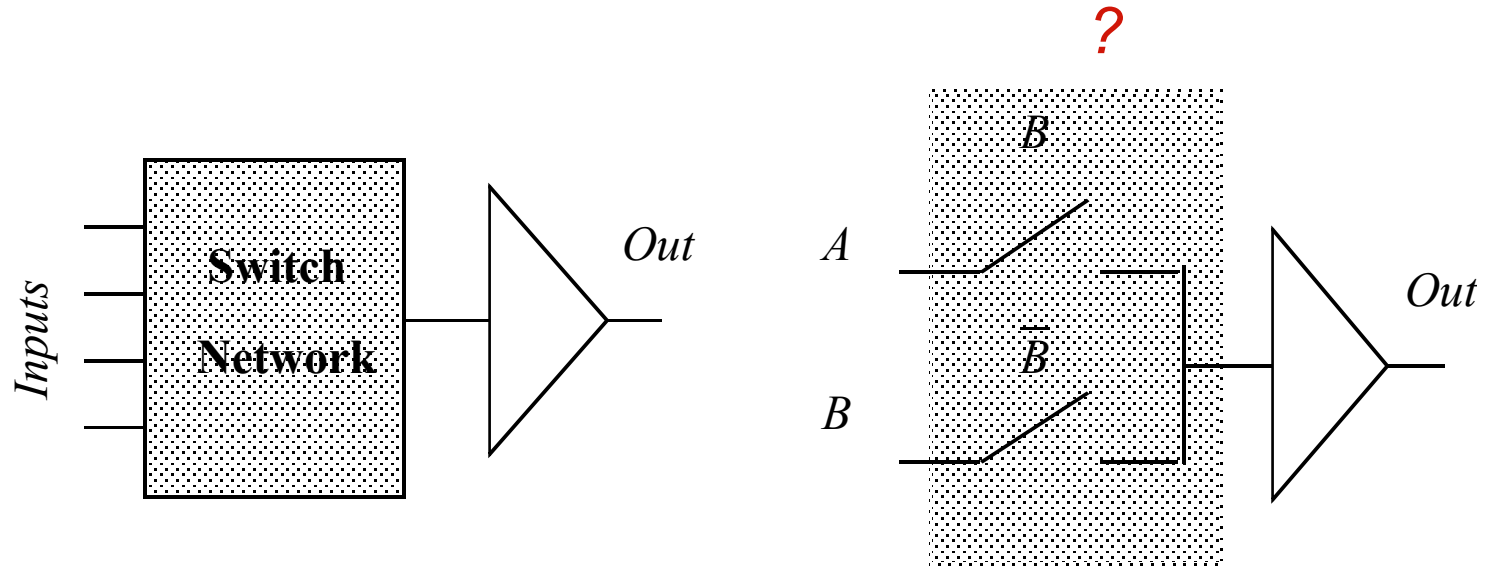
DCVSL Transient Response





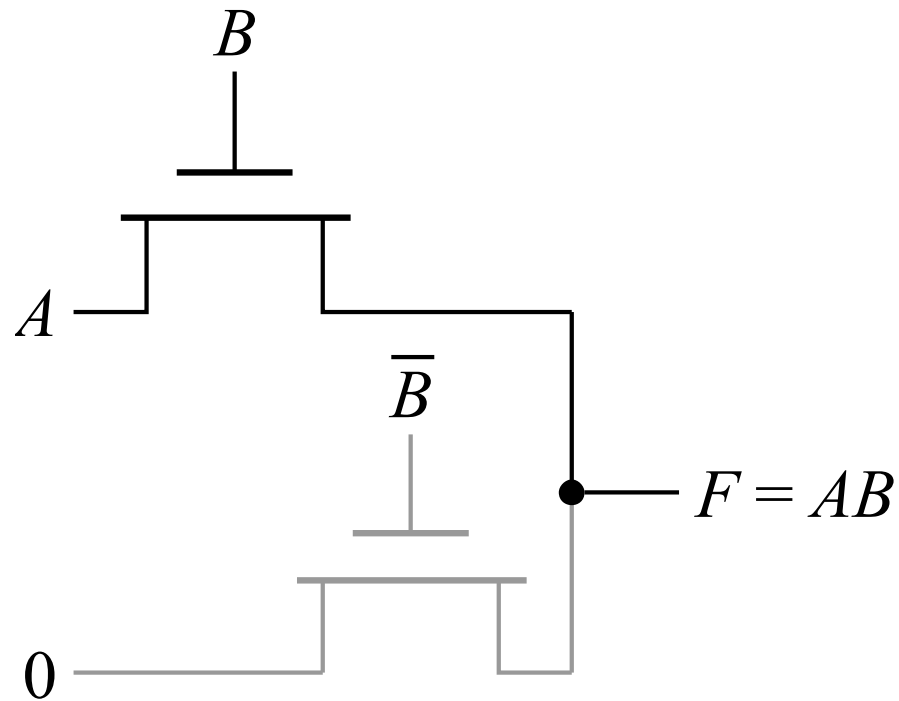
Pass-Transistor Logic

Pass-Transistor Logic

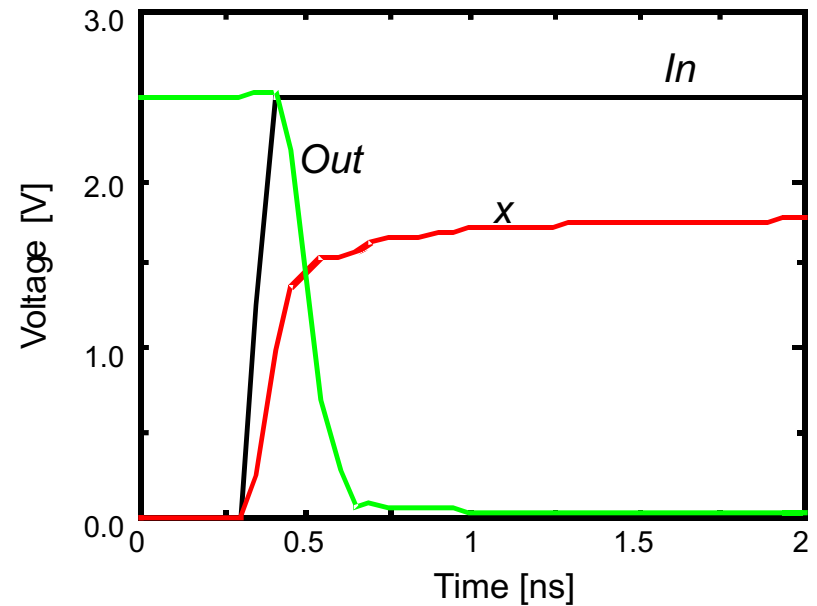
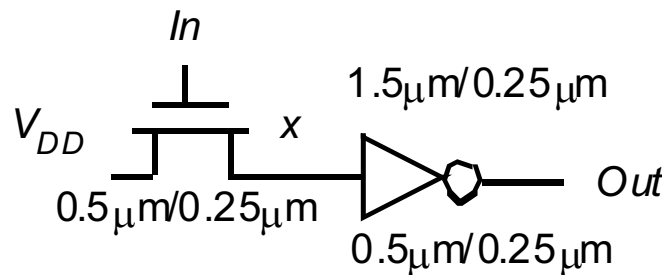


- **N transistors**
- **No static consumption**

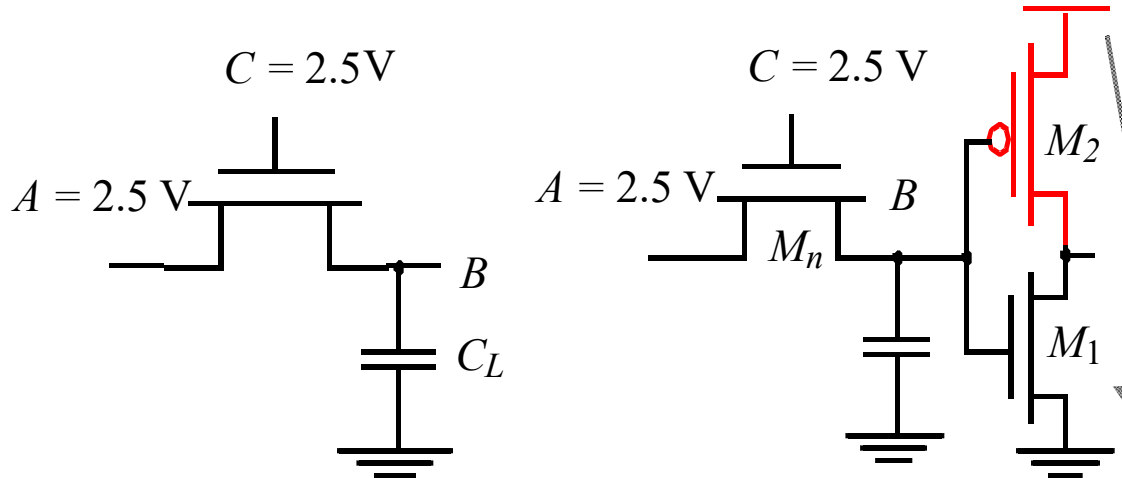
Example: AND Gate



NMOS-Only Logic



NMOS-only Switch

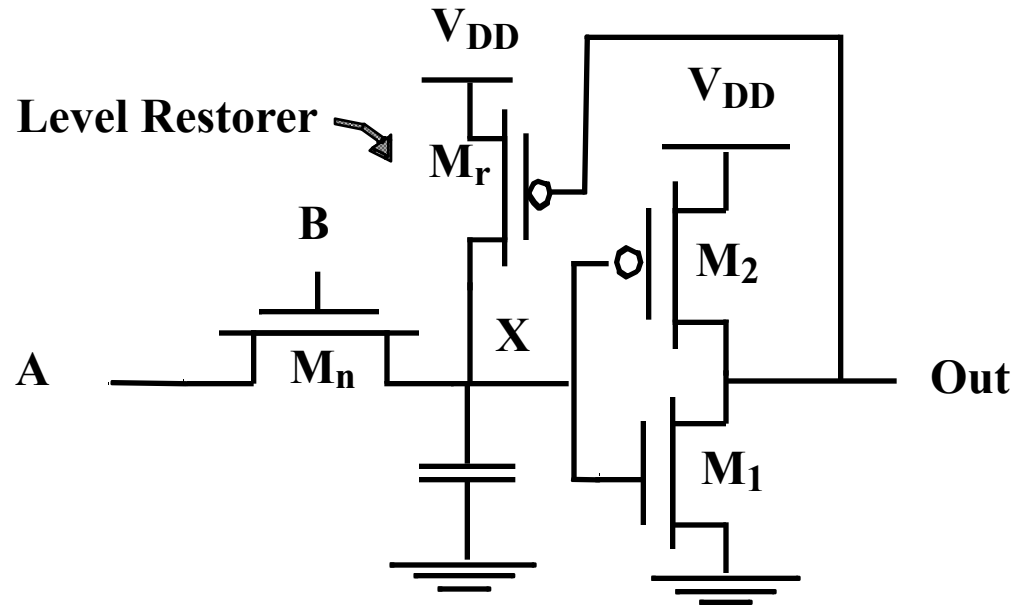


V_B does not pull up to 2.5V, but $2.5V - V_{TN}$

**Threshold voltage loss causes
static power consumption**

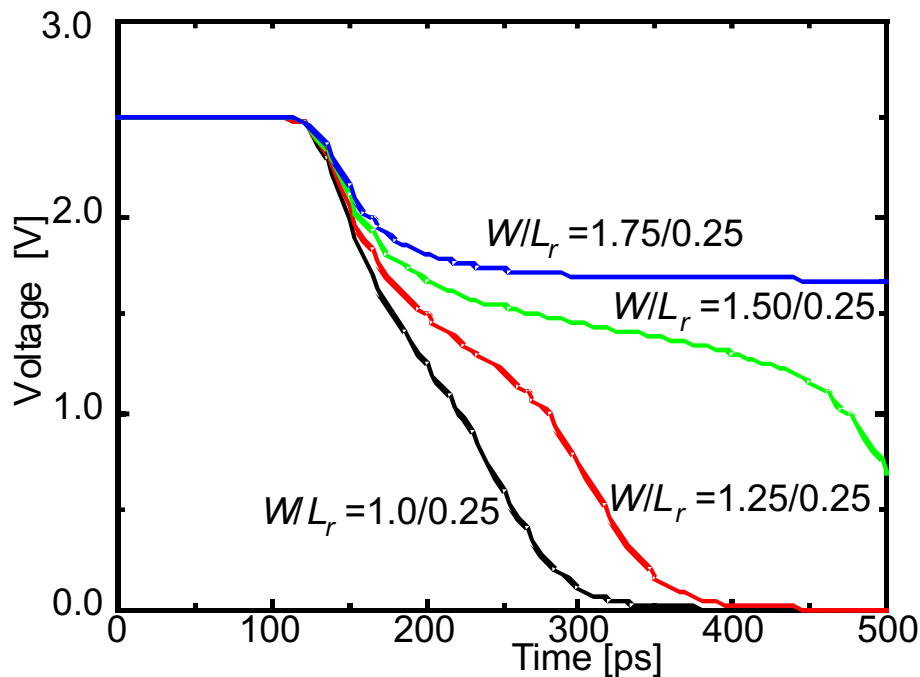
NMOS has higher threshold than PMOS (body effect)

NMOS Only Logic: Level Restoring Transistor



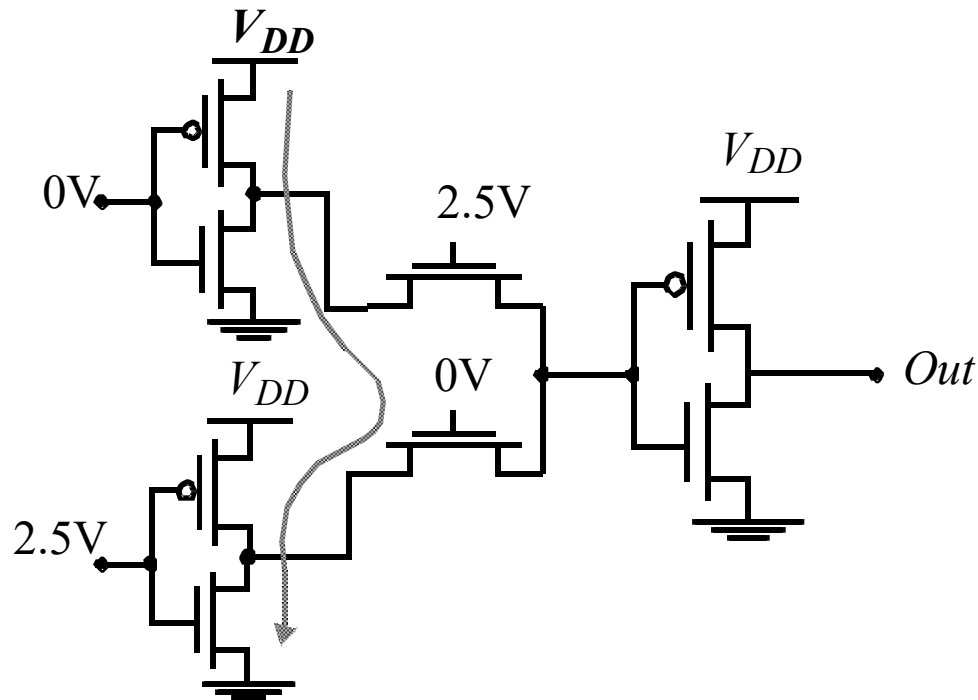
- Advantage: Full Swing
- Restorer adds capacitance, takes away pull-down current at X
- Ratio problem

Restorer Sizing



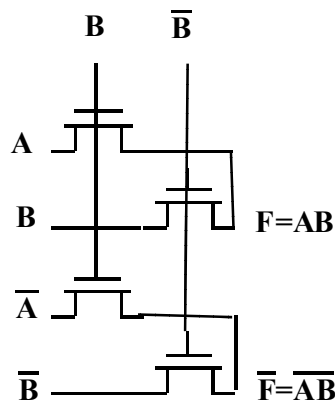
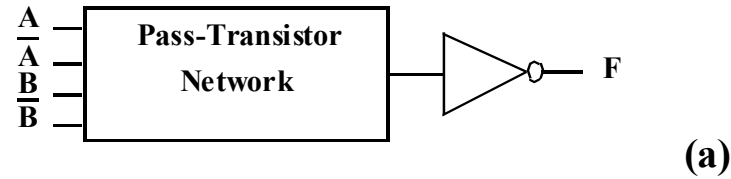
- Upper limit on restorer size
- Pass-transistor pull-down can have several transistors in stack

Solution 2: Single Transistor Pass Gate with $V_T=0$

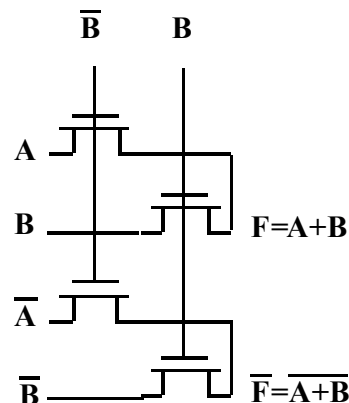


WATCH OUT FOR LEAKAGE CURRENTS

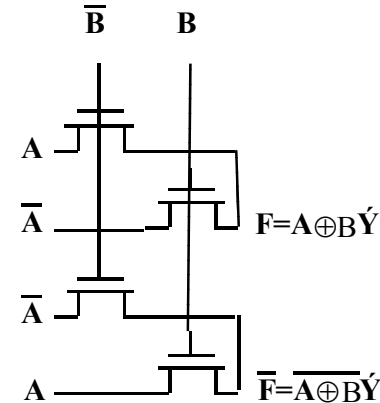
Complementary Pass Transistor Logic



AND/NAND



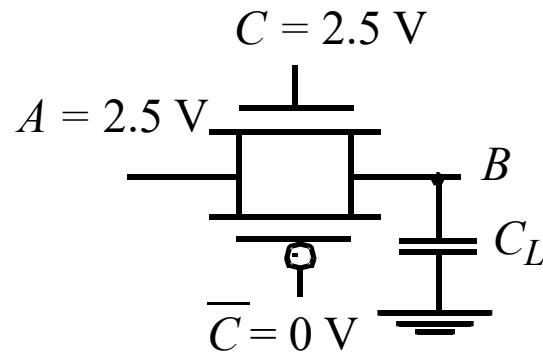
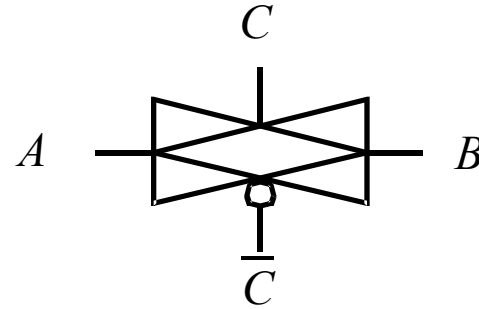
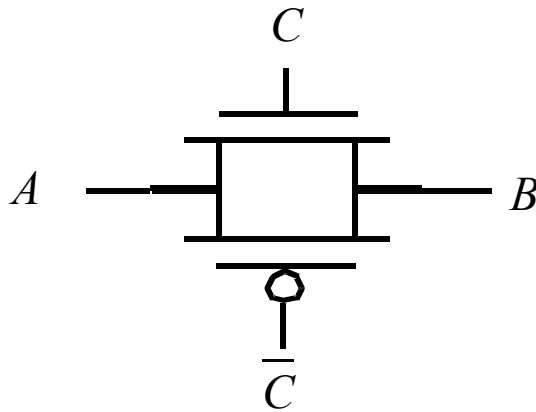
OR/NOR



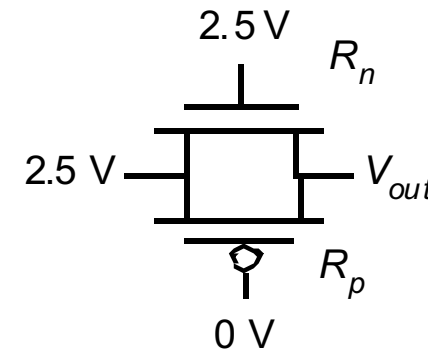
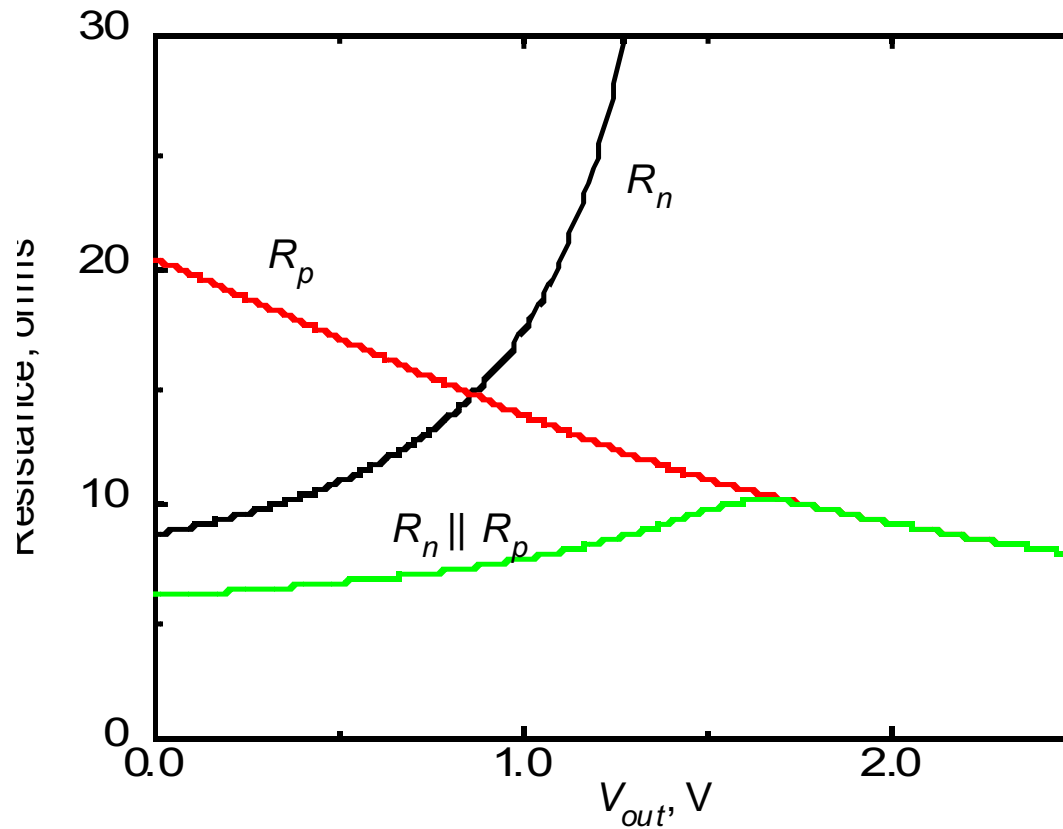
EXOR/NEXOR

(b)

Solution 3: Transmission Gate

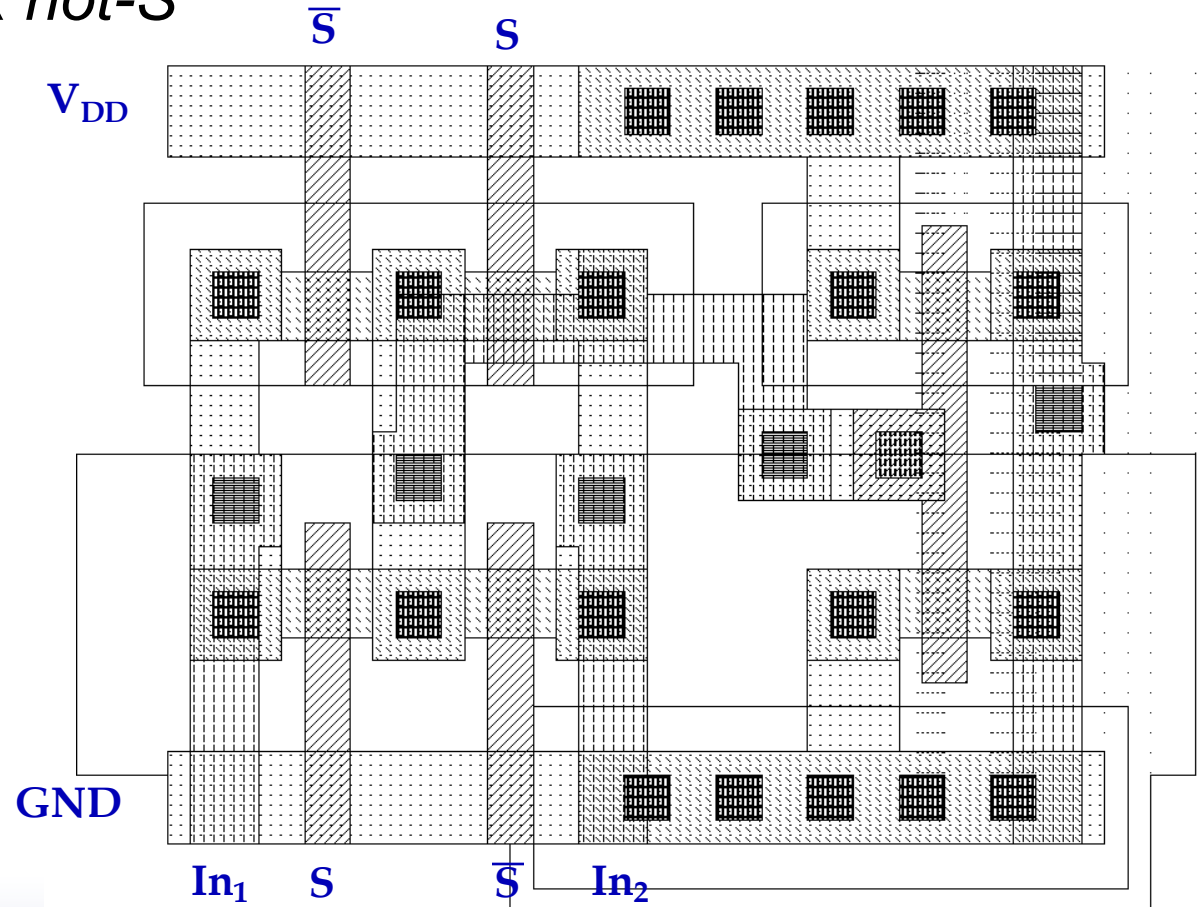
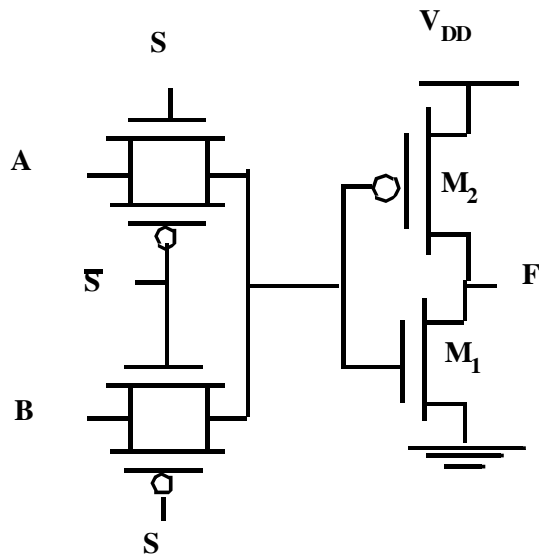


Resistance of Transmission Gate



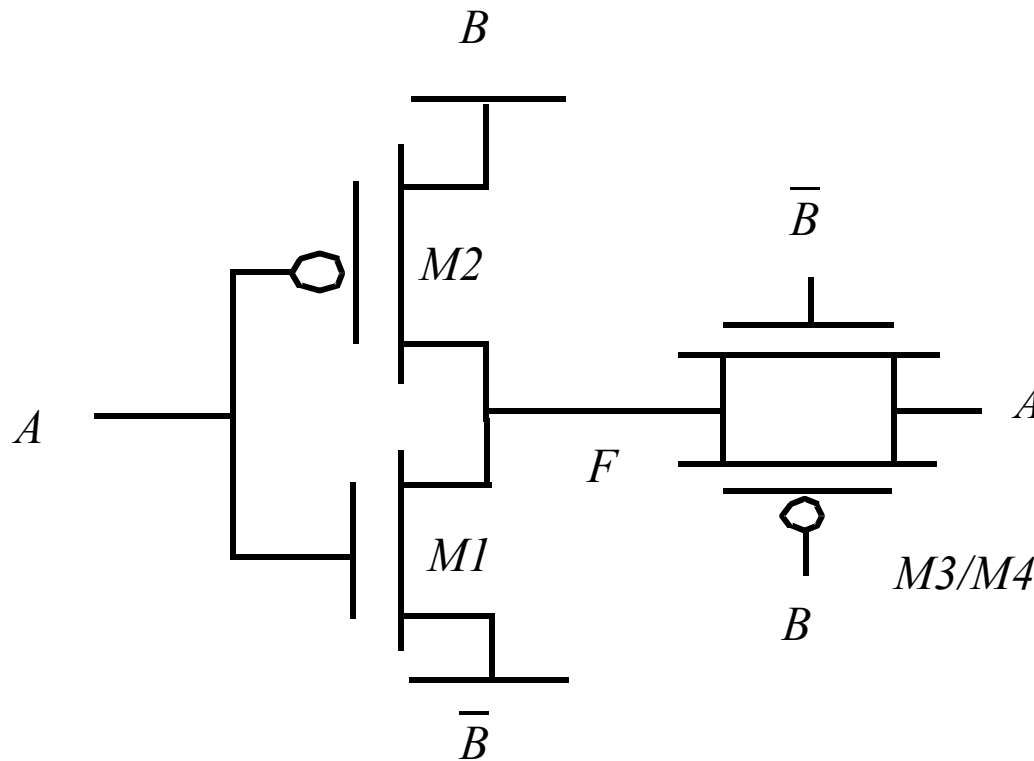
Pass-Transistor Based Multiplexer

$$\text{not-}F = A \times S + B \times \text{not-}S$$

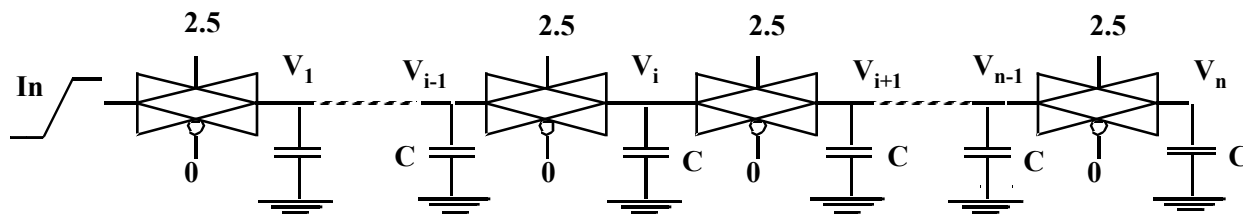


Transmission Gate XOR

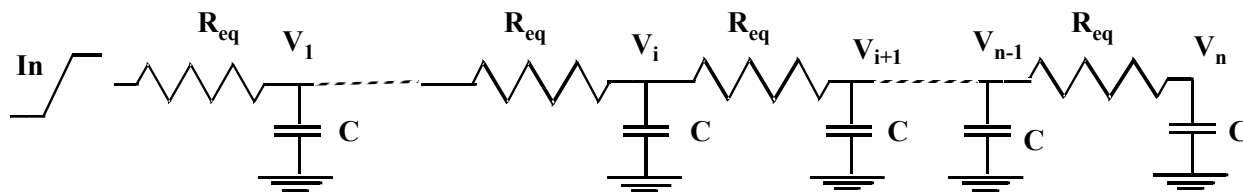
CMOS: 16 MOS



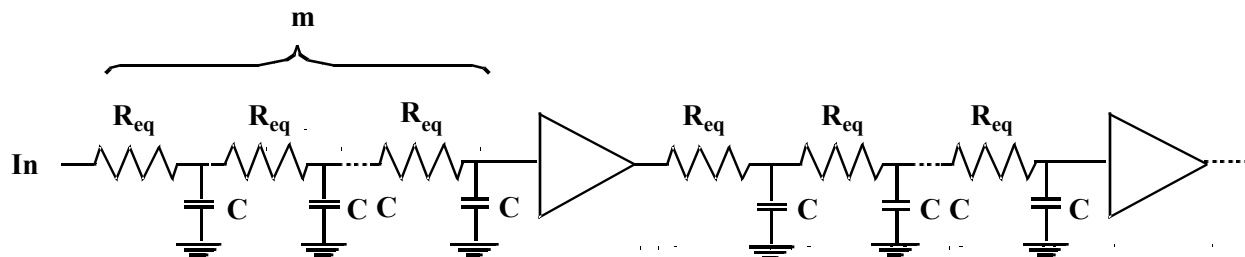
Delay in Transmission Gate Networks



(a)



(b)



(c)



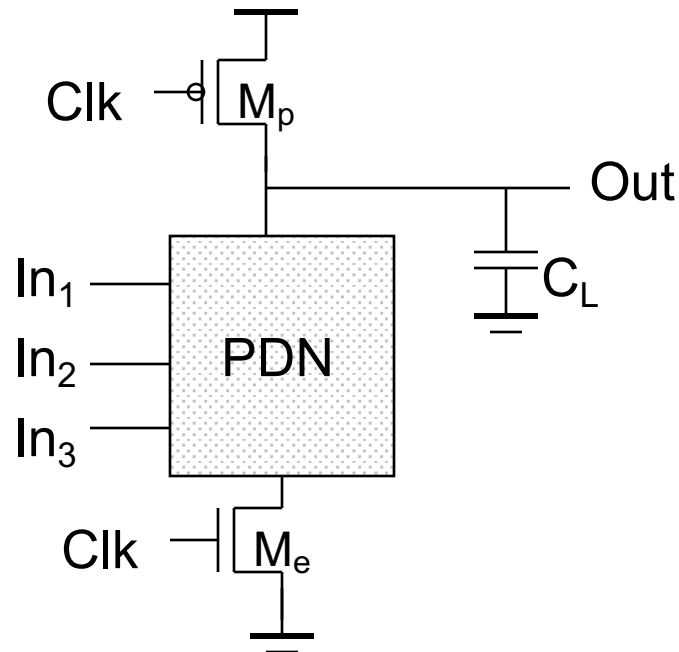
Dynamic Logic

Dynamic CMOS

- ❑ In **static** circuits at every point in time (except when switching) the output is connected to either GND or V_{DD} via a low resistance path.
 - fan-in of n requires $2n$ (n N-type + n P-type) devices

- ❑ **Dynamic** circuits rely on the temporary storage of signal values on the capacitance of high impedance nodes.
 - requires on $n + 2$ ($n+1$ N-type + 1 P-type) transistors

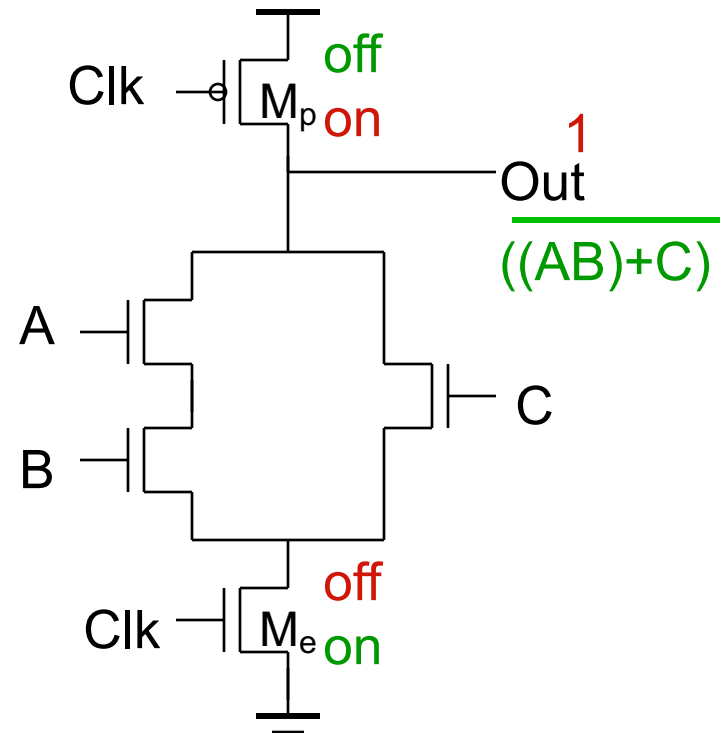
Dynamic Gate



Two phase operation

Precharge (Clk = 0)

Evaluate (Clk = 1)



Evaluate transistor, M_e , eliminates static power consumption

Conditions on Output

- ❑ Once the output of a dynamic gate is discharged, it cannot be charged again until the next precharge operation.
- ❑ Inputs to the gate can make **at most** one transition during evaluation.
- ❑ Output can be in the high impedance state during and after evaluation (PDN off), state is stored on C_L

This behavior is fundamentally different than the static counterpart that always has a low resistance path between the output and one of the power rails.

Properties of Dynamic Gates

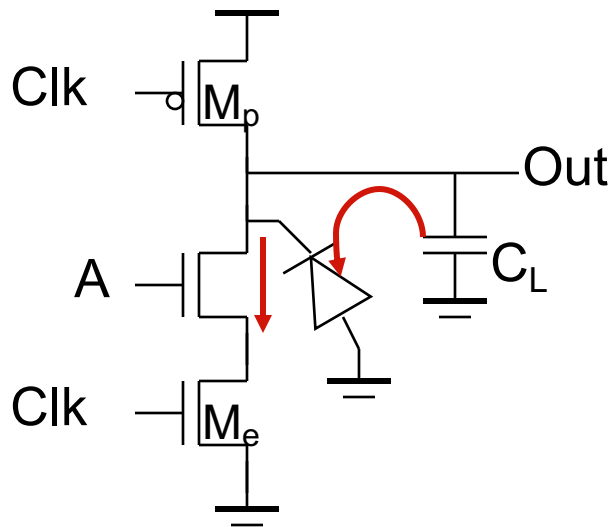
- ❑ Logic function is implemented by the PDN only
 - number of transistors is $N + 2$ (versus $2N$ for static complementary CMOS)
- ❑ Full swing outputs ($V_{OL} = \text{GND}$ and $V_{OH} = V_{DD}$)
- ❑ Non-ratioed - sizing of the devices does not affect the logic levels
- ❑ Faster switching speeds
 - reduced load capacitance due to **lower input** capacitance (C_{in})
 - reduced load capacitance due to smaller output loading (C_{out})
 - no I_{sc} , so all the current provided by PDN goes into discharging C_L

Properties of Dynamic Gates

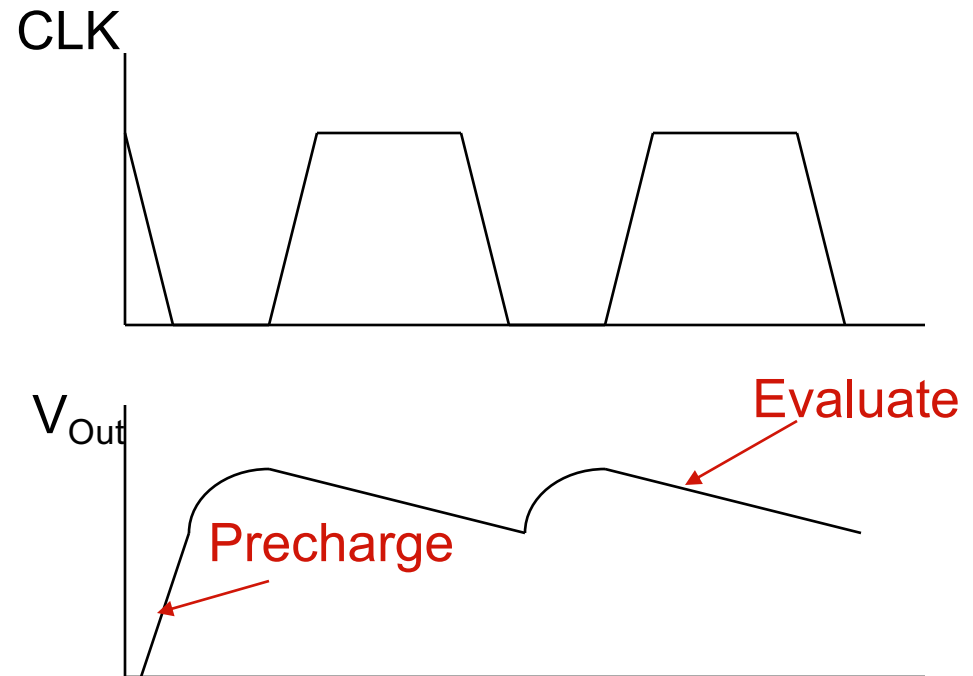
- ❑ Overall power dissipation usually **higher** than static CMOS
 - no static current path ever exists between V_{DD} and GND (including P_{sc})
 - no glitching
 - **higher transition probabilities**
 - **extra load on Clk**
- ❑ PDN starts to work as soon as the input signals exceed V_{Tn} , so V_M , V_{IH} and V_{IL} equal to V_{Tn}
 - low noise margin (NM_L)
- ❑ Needs a precharge/evaluate clock

Issues in Dynamic Design 1: Charge Leakage

F_{min} around 1MHz

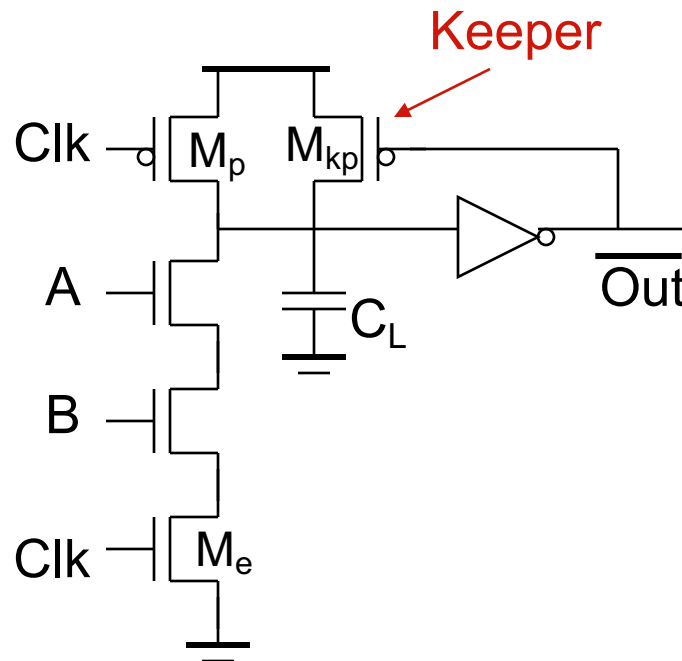


Leakage sources



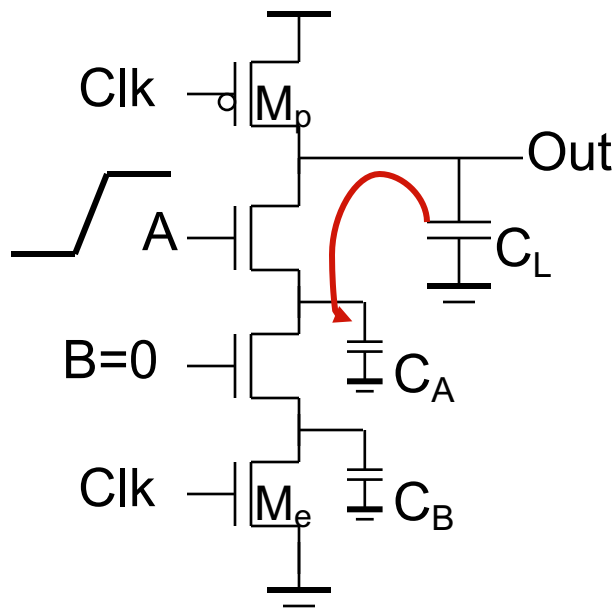
Dominant component is subthreshold current

Solution to Charge Leakage



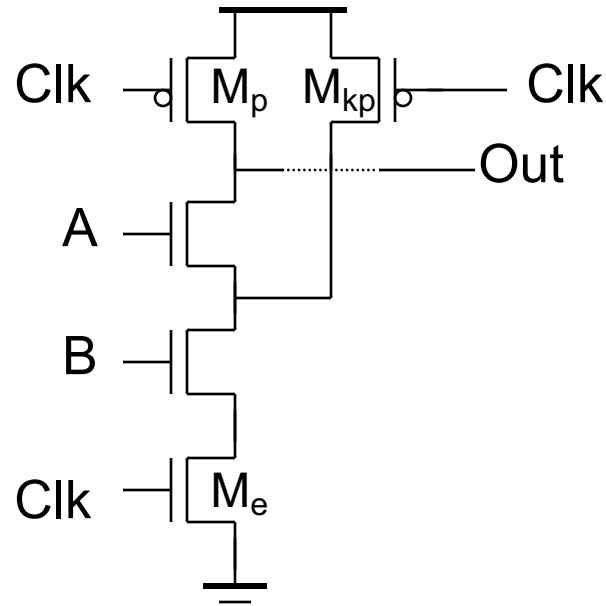
Same approach as level restorer for pass-transistor logic

Issues in Dynamic Design 2: Charge Sharing



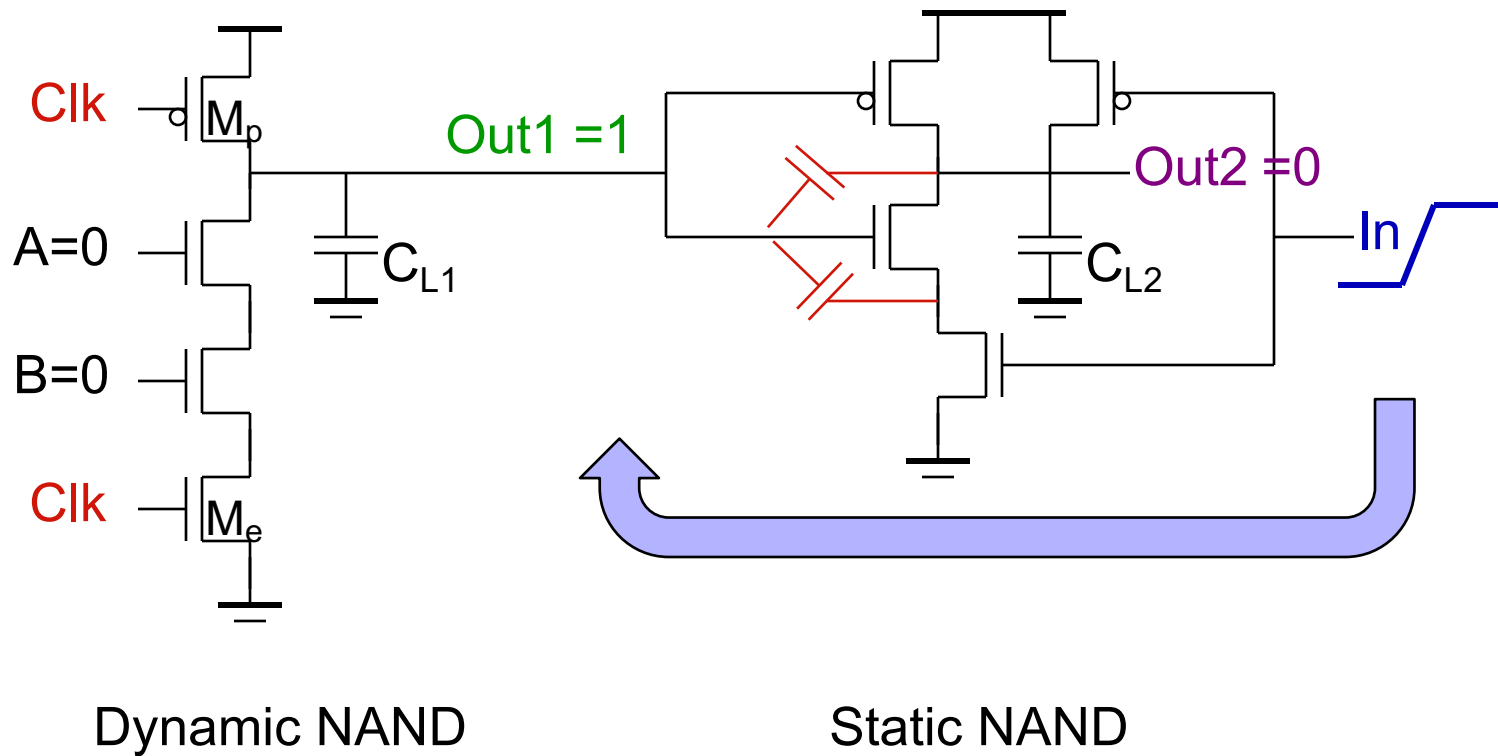
Charge stored originally on C_L is redistributed (shared) over C_L and C_A leading to reduced robustness

Solution to Charge Redistribution

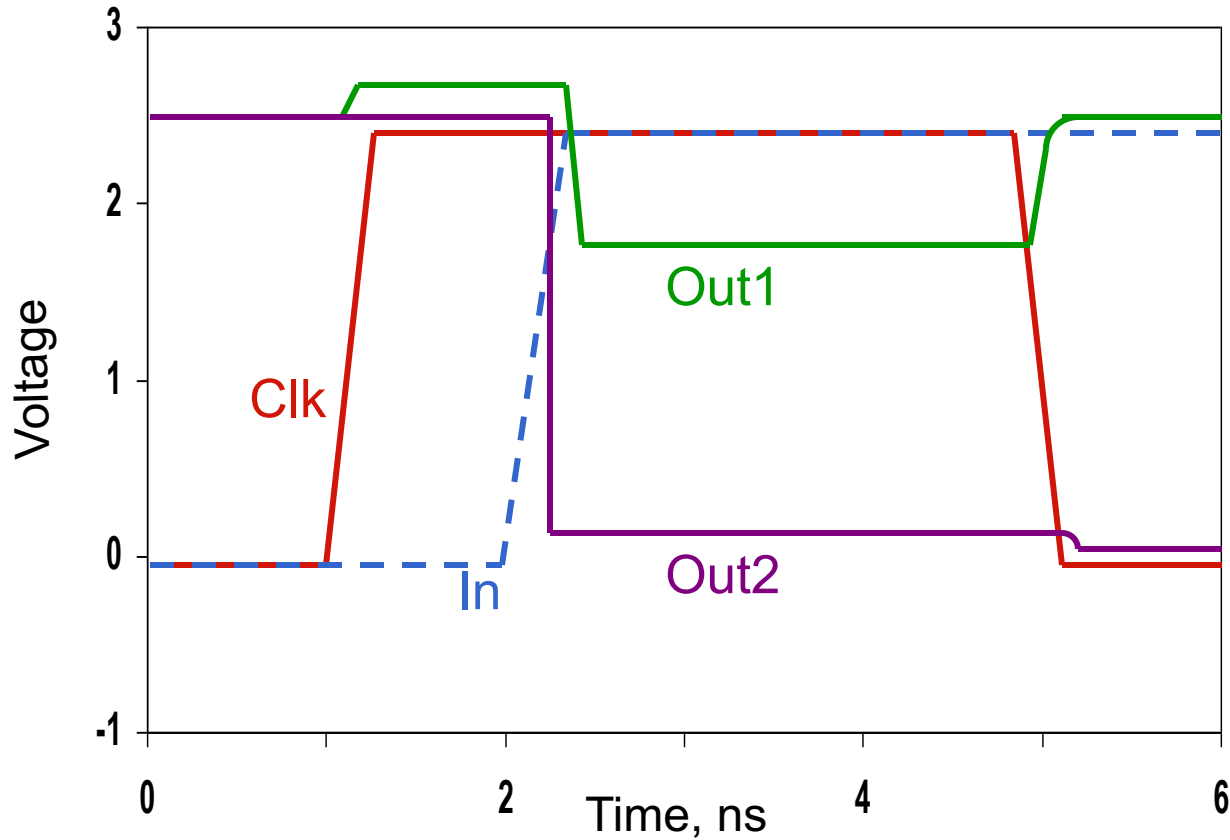


Precharge internal nodes using a clock-driven transistor
(at the cost of increased area and power)

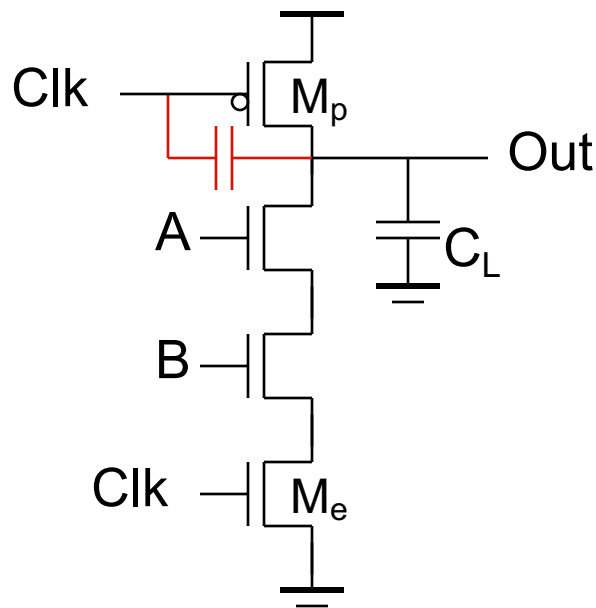
Issues in Dynamic Design 3: Backgate Coupling



Backgate Coupling Effect

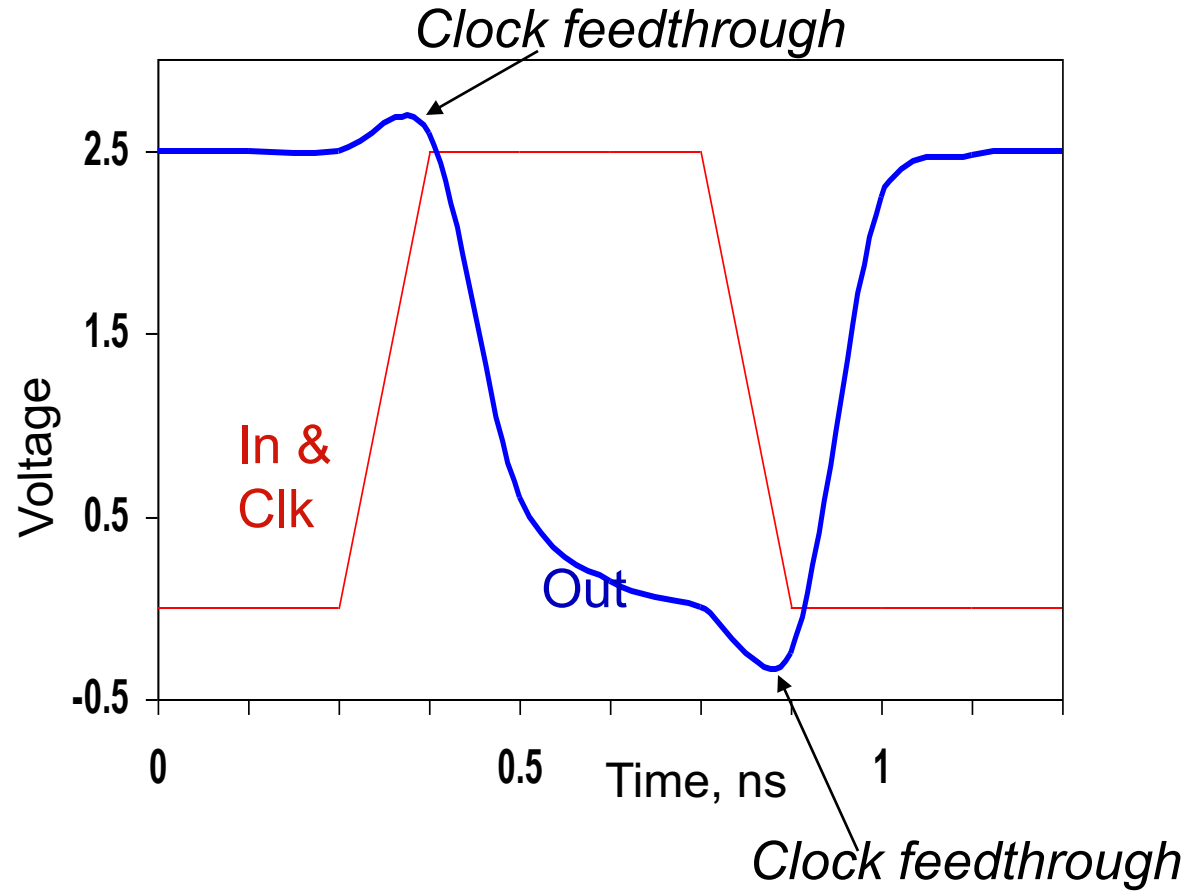
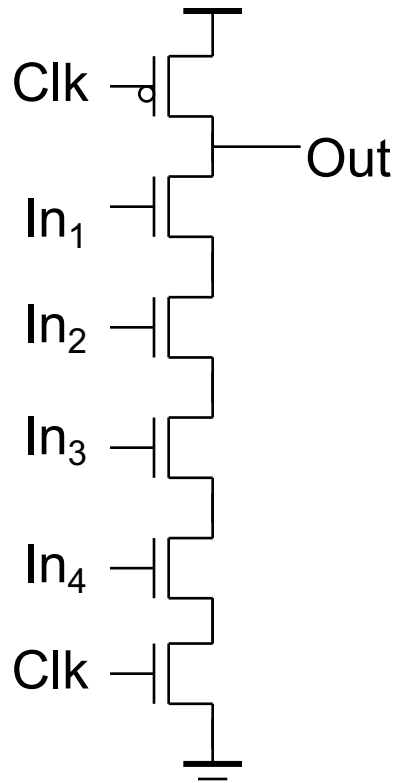


Issues in Dynamic Design 4: Clock Feedthrough



Coupling between Out and Clk input of the pre-charge device due to the gate to drain capacitance. So, voltage of Out can rise above V_{DD} . The fast rising (and falling edges) of the clock **couple** to Out.

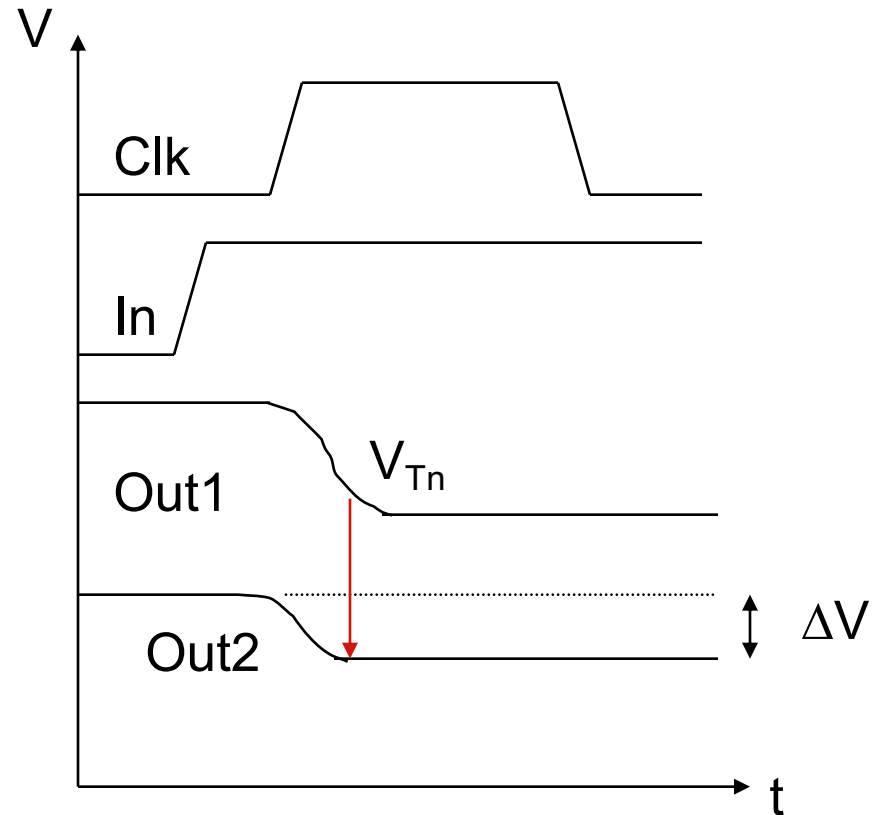
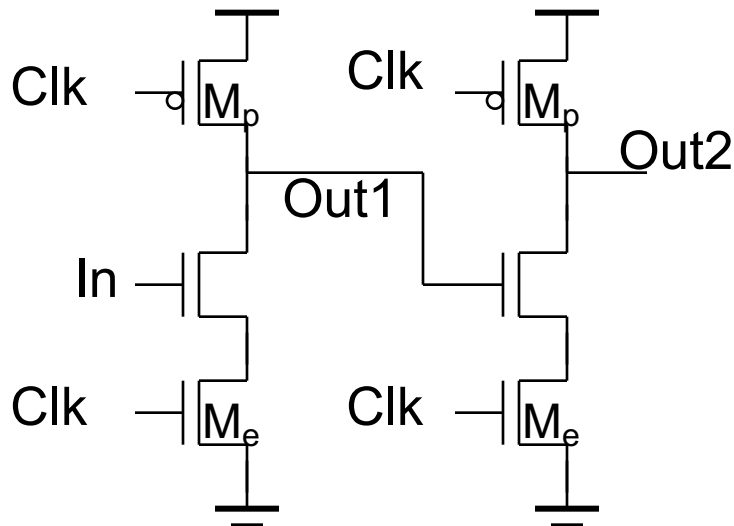
Clock Feedthrough



Other Effects

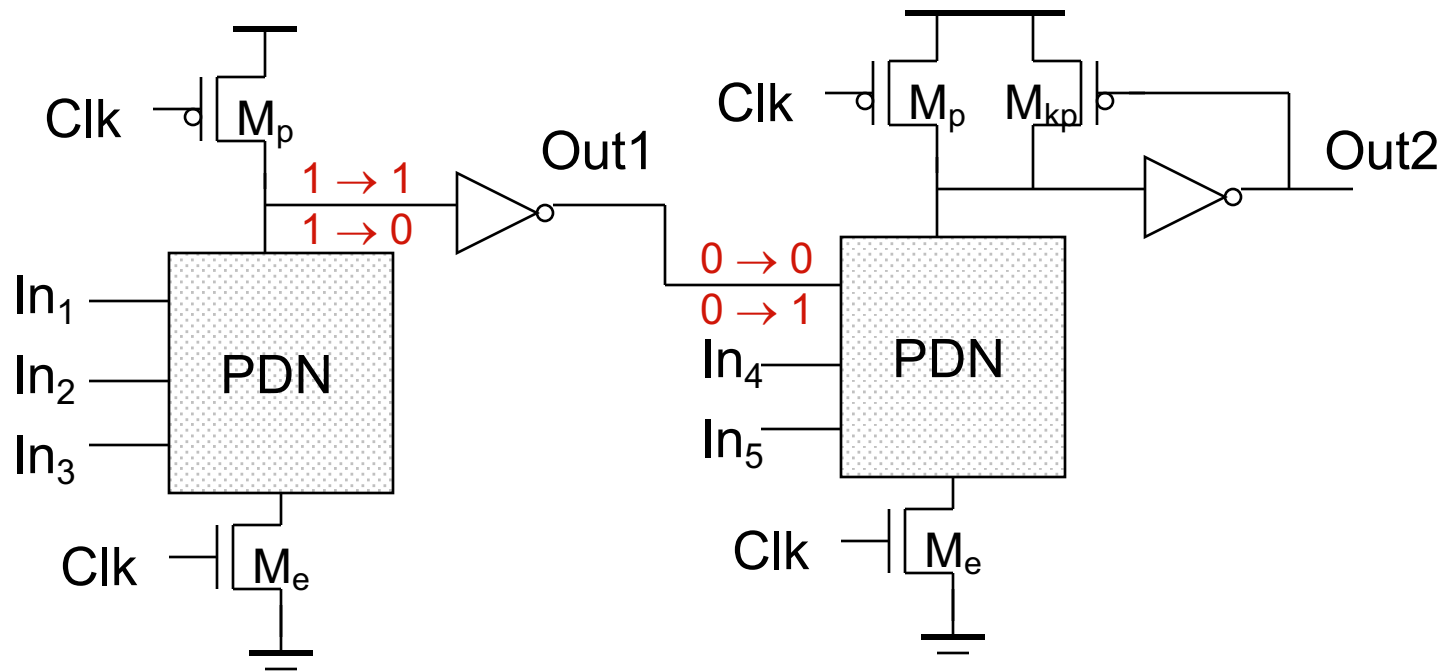
- ❑ Capacitive coupling
- ❑ Substrate coupling
- ❑ Minority charge injection
- ❑ Supply noise (ground bounce)

Cascading Dynamic Gates

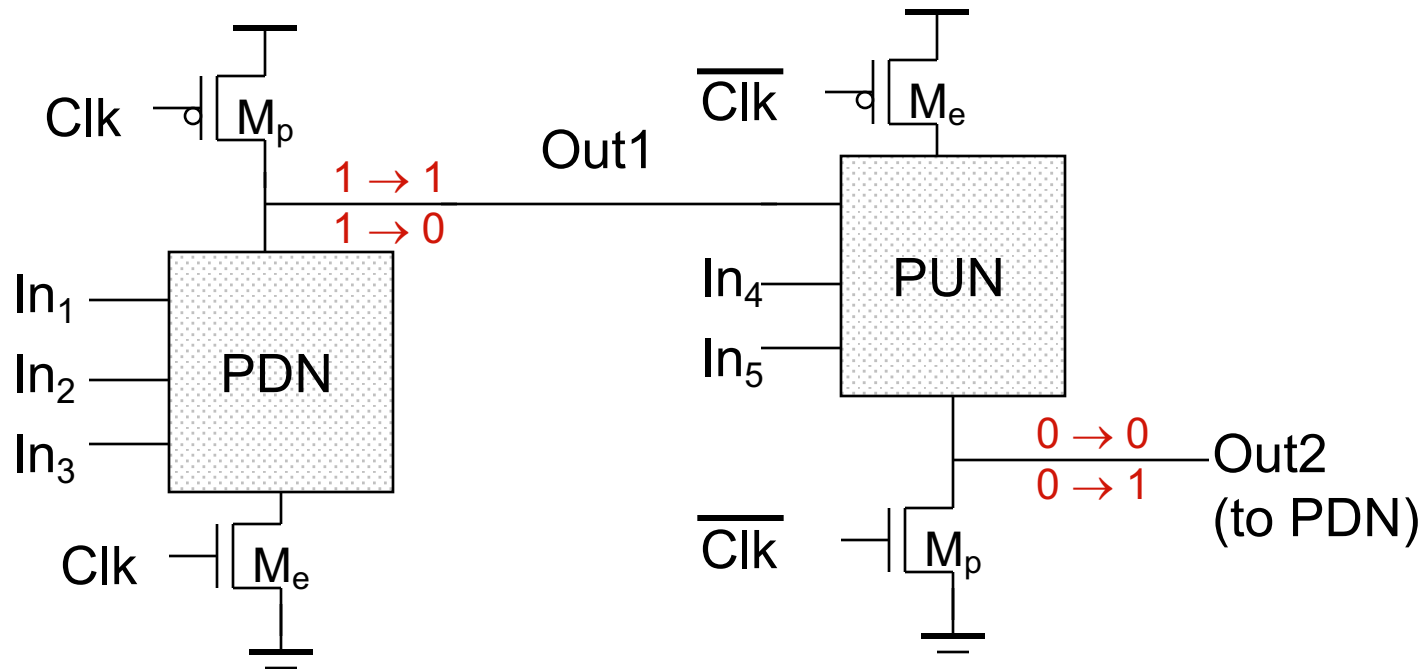


Only 0 $\rightarrow$ 1 transitions allowed at inputs!

Domino Logic



np-CMOS

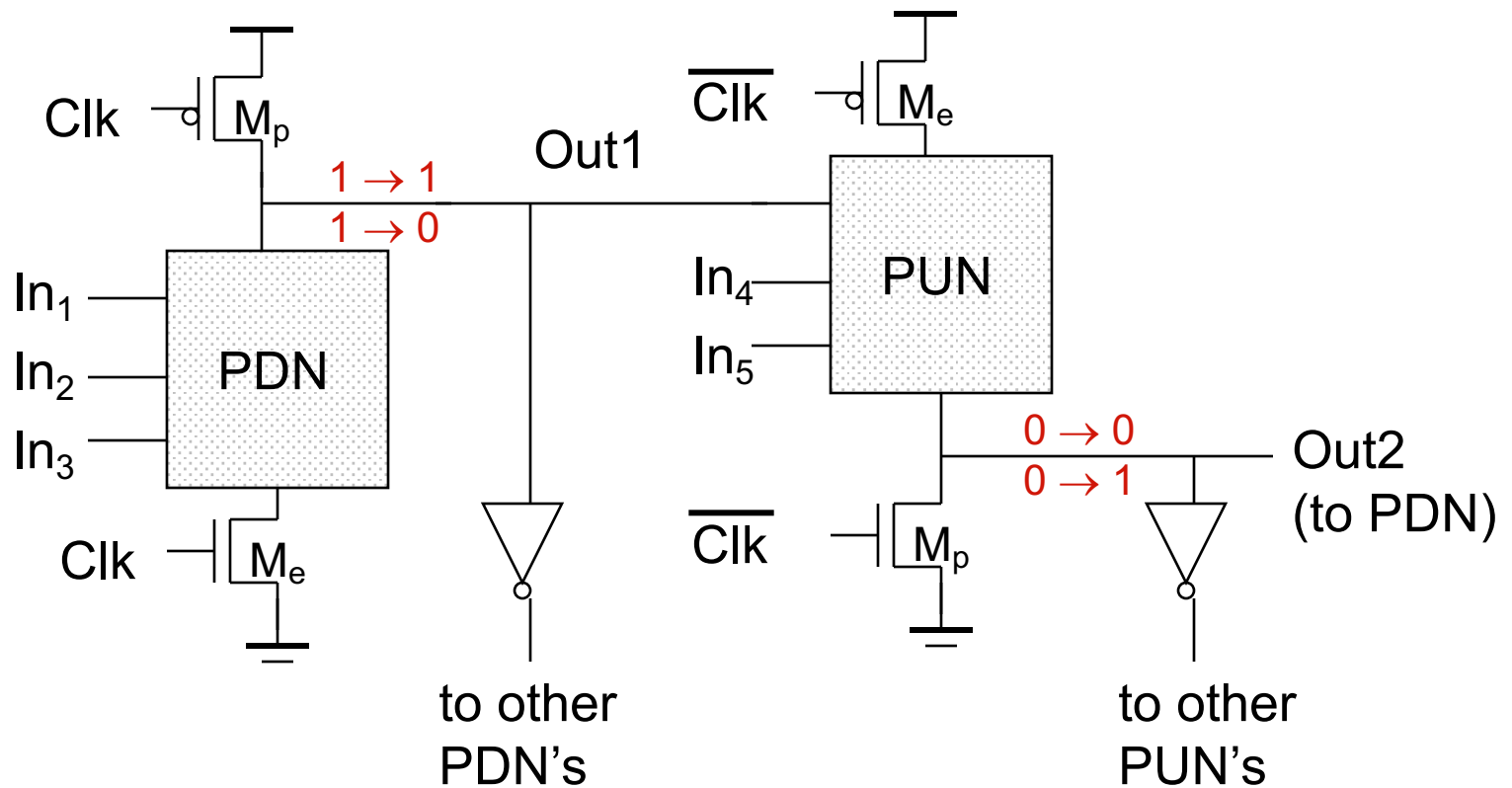


Only $0 \rightarrow 1$ transitions allowed at inputs of PDN

Only $1 \rightarrow 0$ transitions allowed at inputs of PUN

NORA Logic

NORA - no race CMOS



WARNING: Very sensitive to noise!